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EET 4910
Senior Design Proposal

Safe Dip

A Safety-Oriented Hot-Dip Pick-and-Place System

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Abstract

Manual hot-dip coating operations expose workers to elevated temperatures, repetitive strain, and the risk of burns from hot, viscous materials. This project presents Safe Dip, a two-axis Cartesian gantry designed to automate the pick, dip, and place cycle while reducing direct operator exposure. The system provides controlled workpiece positioning, immersion depth, dwell time, and transfer between fixed input, dipping, and output stations.

An STM32 microcontroller coordinates the complete sequence through a cooperative, nonblocking control architecture. Horizontal motion is produced by a stepper-driven timing belt and linear rails, while a stepper-driven leadscrew controls the vertical axis. A conventional electromagnet handles ferromagnetic workpieces, a time-of-flight sensor detects their presence, and normally-closed limit switches provide homing and travel references. Serial and Modbus TCP communications report operating states, sensor information, and fault conditions.

Safety is treated as a primary design consideration. A hardwired Category 0 emergency-stop circuit removes power from the motor drivers independently of the microcontroller, while the electromagnet remains on a separate supply to avoid intentionally releasing a suspended workpiece during a motion stop. The laboratory prototype will be evaluated without a heated coating bath and will demonstrate the integration of the mechanical, electrical, sensing, control, communication, and safety subsystems. Supporting calculations evaluate the design's scalability toward the intended 10 kg payload.

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Chapter 1

Introduction

Summary

In this chapter, we introduce Safe Dip, a safety-oriented hot-dip pick-and-place system designed to remove operators from the hazardous environment of manual wax coating operations. We establish the motivation for automating this process, identify the risks associated with manual dipping, and review existing commercial solutions. We conclude by identifying the gap in the market that Safe Dip is designed to fill.

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1.1 Introduction

Hot-melt dip coating, also referred to as hot dipping or hot-melt wax coating, involves submerging a part into a heated liquid medium such as molten wax, resin, or a similar coating compound. Upon removal, the liquid solidifies as it cools, forming a protective layer around the part that can be cleanly stripped away when needed. This process is commonly applied before long-term storage or bulk packaging to shield parts from corrosion, prevent contact damage that could dull sharp edges, and, in some cases, enhance a tool’s grip or handling comfort [1], [2].

Safe Dip seeks to address this gap by providing an affordable, safety-conscious, and instrumented automation solution tailored to this class of hot-dip operations. It targets the small-to-medium production scale where off-the-shelf industrial robots are cost-prohibitive and general-purpose systems lack the domain-specific safety integration required by industrial electrical standards.

1.2 Motivation

In facilities where this process is still performed manually, workers face sustained exposure to elevated temperatures, repetitive motion injuries, and the risk of burns or chemical contact. As soon as the weight of the objects surpasses a couple of kilograms, long-handled tongs or clippers are introduced, adding to the physical strain. Finally, most coating products are not only hot but also viscous, meaning that even the smallest splash can cause serious injuries. Together, these hazards make manual dipping one of the more accident-prone tasks on the shop floor.

Automating the dipping cycle would remove the operator from the hazard zone entirely, yet many smaller manufacturers have not adopted automated solutions due to cost barriers, system complexity, and the lack of scalable, application-specific equipment suited to their production scale.

In addition to improving operator safety, an effective automation solution should remain useful when the coating process changes. Many commercially available systems are designed around a particular chemical, workpiece geometry, or production workflow, making them difficult or expensive to repurpose. Safe Dip instead treats the dipping liquid as an external process variable: the system performs the same controlled pick, immersion, dwell, and placement sequence regardless of the specific coating, cleaning, or treatment medium being used. This chemical flexibility allows laboratories, prototype facilities, and small manufacturers to adapt to new formulations, suppliers, or applications without redesigning the automation equipment.

By combining this flexibility with a comparatively simple Cartesian architecture, Safe Dip seeks to address the gap between manual dipping and highly specialized industrial machinery. The objective is not to reproduce the throughput or complexity of a complete robotic coating cell, but to provide a lower-cost platform that improves safety, process repeatability, and adaptability at a scale accessible to smaller operations.

1.3 Similar Products

Table 1.1
Comparison of commercial dip coating systems against Safe Dip

Product	Price	Conveyor Feed	Variable Dip Depth	Pick & Place	Chemical Flexibility
Safe Dip (Ours)	~\$500-\$600	YES	YES	YES	YES
Dip-Spin (EFFCO DS-250)	est. \$15k-\$25k	NO	NO	NO	YES
MS800 Tilttable Dip-Spin	est. \$40k-\$80k	NO	NO	NO	YES
Monorail Headstand Screw Machine	\$2,999-\$3,999	YES	NO	NO	Partial
Thread Locker Machine (TH-2004L3)	On Request	YES	NO	NO	Partial

Product	Price	Conveyor Feed	Variable Dip Depth	Pick & Place	Chemical Flexibility
Full Auto Thread Coater	\$20k–\$30k	YES	NO	NO	NO
Emmeti WAX-ON	est. \$50k–\$150k+	YES	YES	YES	Partial
South Shore Robotic Dip Cell	Custom Quote	NO	YES	YES	YES

Chemical flexibility is the ability of a system to operate with different coating, cleaning, and treatment liquids without requiring changes to its mechanical architecture or control logic. The system interacts with the process only through part detection, controlled handling, programmable immersion, and placement; therefore, the chemical composition of the liquid is treated as an external process variable rather than a machine-specific design input. Compatibility is limited primarily by the temperature tolerance and chemical resistance of the end effector and other exposed hardware, rather than by the operating principle of the system itself.

This separation between the machine and the dipping chemistry is important because it allows the same platform to be repurposed as process requirements, suppliers, or formulations change. Instead of designing dedicated equipment for each liquid or application, users can adapt the process by changing the dip medium and operating parameters while retaining the same automation hardware. This reduces dependence on a single chemical process, lowers the risk of future redesign, and makes Safe Dip more suitable for laboratories, prototyping environments, and small-scale manufacturing where process requirements may change frequently.

1.3.1 Aramfix EFFCO DS-250 Dip-Spin Machine

The EFFCO DS-250 is a batch-type industrial dip-spin coating machine designed for high-volume processing of small metal components such as fasteners, nuts, bolts, and stampings. The machine operates by loading parts into a perforated basket, submerging the entire basket into a coating tank, and then spinning the basket at high RPM to centrifugally remove excess

coating and achieve a uniform film thickness. This process enables extremely consistent results across thousands of parts per cycle, making it well-suited for mass production environments. The DS-250 features a heated coating tank and programmable dip and spin cycles, allowing operators to fine-tune coating thickness to specification. However, its batch-based design means individual part handling and positional control are not possible, as all parts in the basket are processed simultaneously without differentiation. The machine requires manual loading and unloading of the basket, meaning human intervention is necessary at each production cycle. Its industrial footprint and price range of \$15,000 to \$25,000 place it well beyond the reach of small-scale or research-oriented operations. Unlike Safe Dip, the DS-250 cannot adapt to varying part geometries within a single run or interact with a conveyor-based workflow.

1.3.2 MS800 Tilttable Dip-Spin Machine

The MS800 is a fully automatic, tilttable dip-spin coating machine designed for heavy-duty industrial production of fasteners and small precision components. Its distinguishing feature is a motorized tilting mechanism that repositions the basket during the drain cycle, improving coating drainage and reducing waste compared to standard vertical-axis spin machines. The system is PLC-controlled, offering precise regulation of dip duration, spin speed, tilt angle, and temperature, resulting in coating uniformity that is difficult to match with manual or semi-automatic methods. With a throughput capacity in the thousands of parts per hour, the MS800 is optimized for large-scale manufacturing lines rather than flexible or small-batch applications. The machine's price range of \$40,000 to \$80,000 reflects its industrial-grade construction and automation level. Like other dip-spin systems, it is fundamentally a batch processor and cannot handle individual components, vary dip depth per part, or interface with a pick-and-place conveyor system. Its complexity also demands trained operators and dedicated maintenance protocols. While it represents the high end of coating machine capability, it addresses a fundamentally different operational context than Safe Dip.

1.3.3 Monorail Headstand Screw Coating Machine

The monorail headstand screw coating machine is a continuous-flow system designed specifically for the automated surface treatment of headed fasteners such as screws, bolts, and rivets. Parts are loaded onto an overhead rail conveyor by their heads, hanging inverted as they travel through a dip station where their threaded sections are coated in the desired chemical or lubricant. The continuous rail feed allows throughput rates of 10,000 to 20,000 parts per hour, making it one of the most efficient solutions for large-volume screw coating applications. Because parts self-hang from their heads, no gripper or pick-and-place mechanism is

required, which simplifies the mechanical design significantly. However, this same feature makes it incompatible with any component that lacks a head or flange to hang from, limiting its applicability to a narrow category of fastener geometry. The coating type per production run is fixed, and changing the chemical requires a full tank flush and reconfiguration. Priced between \$2,999 and \$3,999, it is the most affordable automated option reviewed, but its single-purpose nature and geometric constraints make it unsuitable for the diverse component handling that Safe Dip is designed to support.

1.3.4 Thread Locker Coating Machine — TH-2004L3

The TH-2004L3 is a purpose-built automatic dispensing machine designed to apply thread-locking adhesive, such as Loctite or Vibra-Seal equivalents, directly onto the threads of screws and bolts prior to assembly. The machine integrates a bowl feeder for bulk part loading, a precision dispensing head that applies a metered quantity of adhesive to each thread zone, and a conveyor exit for processed parts. Its dispensing mechanism is calibrated to apply only to the thread region, avoiding contamination of the head or shank, which is critical for assembly line compatibility. The system handles fastener sizes from M3 to M36 and operates at high throughput, making it appropriate for automotive, electronics, and aerospace assembly environments. Priced between \$3,300 and \$4,500, it sits at the lower end of the industrial machine spectrum, though it is still an order of magnitude more expensive than Safe Dip. Its fundamental limitation is that it is chemically and geometrically constrained: it applies a single fluid type to a single class of headed fasteners, with no capability to handle other part geometries or coating materials. It also has no pick-and-place functionality and cannot interface with a generalized conveyor workflow.

1.3.5 Full Automatic Thread Coater

The full automatic thread coater is a high-capacity, end-to-end automated machine designed for the uninterrupted production-line application of thread-locking compounds or sealants on screws and bolts. It integrates automated part feeding, precision coating application, and a discharge conveyor into a single self-contained unit, eliminating the need for manual intervention during normal operation. The system is optimized for single-fluid, single-geometry workflows and is typically configured for a specific fastener size range at installation. Throughput is very high, and the machine is engineered for continuous multi-shift operation in demanding industrial environments. Priced between \$20,000 and \$30,000, it represents a significant capital investment best justified in high-volume production contexts. Its automation level, while impressive within its domain, comes at the cost of flexibility: changing the coating fluid or accommodating a different part geometry requires machine reconfiguration

and downtime. It lacks any conveyor-integrated pick-and-place capability and cannot adjust dip depth or dwell time per individual part. These constraints make it an effective solution for its intended use case but a poor match for the adaptive, multi-geometry coating scenario that Safe Dip is designed to address.

1.3.6 Emmeti WAX-ON Machine

The Emmeti WAX-ON is a fully robotic, end-to-end bottle waxing line developed for high-volume industrial winery and spirits production. The system integrates a depalletizer, an infeed conveyor, a robotic dipping arm, a multi-tank wax station with interchangeable color tanks on rails, and a vision-based defect inspection camera, all feeding into an automated case packer. Its robotic arm performs true pick-and-place operations, selecting individual bottles from the conveyor and submerging them into the wax tank with consistent depth and dwell time control. The 3D part identification system allows the machine to adapt to minor variations in bottle positioning without human correction. Coating uniformity is achieved through a combination of precise dip speed, controlled withdrawal rate, and optional rotation of the bottle during extraction. The system is capable of processing hundreds of bottles per hour with minimal operator involvement, making it one of the most capable waxing machines commercially available. However, its price range of \$50,000 to \$150,000 and its large industrial footprint place it entirely outside the scope of small-scale or research applications. It is also purpose-designed for bottles and is not adaptable to other component geometries without significant engineering modification.

1.3.7 South Shore Controls Robotic Dip Cell

The South Shore Controls robotic dip cell is a custom-engineered industrial automation solution designed for precision dipping, coating, and chemical treatment of components across a range of industries, including food processing, medical device manufacturing, and industrial finishing. The system is built around a 6-degree-of-freedom industrial robot arm that provides full spatial control over the component throughout the dip cycle, enabling complex dip trajectories, variable dip angles, and multi-stage coating sequences that rigid gantry systems cannot replicate. Custom end-effectors, or grippers, are designed to match the specific geometry of the target component, ensuring reliable handling even for irregular or delicate parts. The system is PLC-controlled and fully programmable, with PID-regulated temperature control on the coating tank and real-time process monitoring. Pricing is entirely quote-based and typically falls in the range of \$50,000 to \$100,000 depending on configuration, making it accessible only to well-capitalized industrial clients. Unlike Safe Dip, it does not incorporate a conveyor-based intake and exit workflow; parts must be presented to the robot

by separate handling systems. Its strength lies in precision and flexibility within a controlled cell environment, rather than inline conveyor integration.

1.4 Other Alternatives

Before arriving at the proposed design, several alternative approaches were evaluated across three key areas of the system: the overall mechanical architecture, the end effector mechanism, and the fail-safe strategy for power loss events. Each alternative was assessed against the project's core priorities of operator safety, cost, mechanical simplicity, and suitability for the hot-dip environment. The following subsections document the options considered and the reasoning behind the decisions made.

1.4.1 Mechanical Architecture

From the very beginning of the project, the idea of a gantry competed against the idea of a robotic arm, as the one in Fig. 1.1. While it presented dexterity enough for the task at hand, a couple of issues remained, regardless of the degrees of freedom it presented:

- The bigger the number of degrees of freedom it had, the bigger the electronic and computer complexity the system would carry, as each degree of freedom would require at least an additional actuator.
- Mechanically, the fact that the arm is only anchored at one point at the bottom, offset from the object's position, meant that the motors would have to generate and support the entirety of the weight of the object.
- Given the height of some hot-dipping pots, the arm would either require unusually long sections, which would toll even more the motors to sustain the torque, all this to be able to get deep enough for the pot, and high enough to avoid it while grabbing the object.

In contrast, the mechanics of a cartesian gantry are pretty straightforward: only a motor per axis is required, and the positioning of them makes it such that the structure bears most of the weight in a predictable way.

Finally, a last alternative was considered where the objects being dipped could just move through a conveyor belt and have the liquid poured from the top, but that was not acceptable, since materials like hot wax would harden and break the belt, and a uniform coating would be hard to achieve.



Fig. 1.1: Universal Robots UR5e Collaborative Robot Arm

1.4.2 Effector Alternatives

Another aspect worth consideration is the end effector (i.e., the mechanism that grabs the objects). While a claw is usually the first option everyone thinks of, it also comes with its own disadvantages:

- Claws must have a strong enough grip to ensure the object doesn't slip, while being soft enough as to not deform it or break it.
- The grip strength is dependent on the geometry of the object and the positioning of the claw.
- As the weight of the objects to be picked increases, it's been recommended to use a pneumatic claw, which introduces a new design dimension, complications, and considerations.

While an electro-permanent magnet would have been a more suitable effector, more aligned with the safety-conscious objective of this project, its availability is limited, and its price is cost-prohibitive for a prototype.

1.4.3 E-Stop Alternatives

Finally, for a project created out of the need for a safe alternative to manual operation, a lot of thought has been put into how the system should behave in the case an emergency stop was required. International standard IEC 60204-1 specifies three categories of emergency stops [3]:

- Category 0: uncontrolled stop by immediate removal of actuator power.
- Category 1: controlled stop, with power retained long enough to stop, followed by power removal.
- Category 2: controlled stop with actuator power remaining available.

Category 1 requires more complexity than category 0, since it requires hardware to disconnect power after a specific time. While stepper motors could benefit from a soft deceleration, given the speed they reach, the complexity outweighed the benefits for this project. Furthermore, the microcontroller cannot be used to trigger a power-off sequence; it is always assumed that it can fail too.

Category 2 simply didn't fit the requirements of this project.

Chapter 2

Proposed Work

Summary

This chapter describes the proposed design of Safe Dip in detail. We explain the system's two-axis Cartesian gantry architecture and define its operating limitations. We present the testable system requirements and their verification criteria, describe each functional module through a system block diagram, provide engineering specifications for each subsystem, and outline the project budget.

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2.1 Overview of Safe Dip

Safe Dip is a two-axis Cartesian gantry system designed to automate the pick, dip, and place cycle of a hot-melt coating operation. The horizontal axis is driven by a timing belt coupled to a stepper motor along a pair of linear rails, providing smooth and repeatable travel between the input conveyor, the dipping pot, and the output conveyor. The vertical axis is driven by a leadscrew, which is mechanically self-locking, based on its pitch and the workpiece's weight, meaning that the carriage holds its position without active braking in the event of a power loss or fault. The lead nut is mounted directly to the horizontal slider, so the entire vertical assembly raises and lowers as a unit, keeping the gantry clear of the dipping pot during horizontal travel. Together, these two axes execute a coordinated motion sequence that lifts parts from an input position, lowers them into the heated coating bath for a controlled dwell period, withdraws them, and deposits them at an output position, removing the operator from direct contact with the hazardous thermal environment.

The prototype operates at a reduced payload and process temperature relative to the full industrial specification in order to meet laboratory safety requirements. The control architecture, communications stack, and safety system topology are nonetheless implemented at full fidelity, and design calculations provided in this report demonstrate how the system scales to the industrial target.

2.2 Operating Limitations

The following constraints define the boundaries within which Safe Dip is designed to operate, which are outside the control of the system. They are not subject to pass/fail testing, but scope the design and inform component selection.

- Safe Dip operates along two axes only: a horizontal axis for transfer between the input station, dipping bath, and output station, and a vertical axis for picking, dipping, and placing the workpiece. The system does not support rotation or movement perpendicular to the horizontal travel path.
- The system is designed to handle cylindrical workpieces within the specified size and payload limits.
- Workpieces must be presented at a known, fixed input position. The system does not perform lateral alignment, orientation correction, or vision-based localization.
- Each workpiece must provide a clean, flat, ferromagnetic attachment surface that remains outside the required coating area. Because the electromagnet remains in contact with this surface throughout the cycle, the attachment area will not be dipped or coated. Workpieces

requiring complete surface coverage would require a different end effector or a secondary coating operation.

For the laboratory prototype, the following additional operating limitations apply:

- The laboratory prototype is designed and validated for workpieces with a maximum mass of 1 lb. Operation above this payload is outside the prototype's validated operating range.
- The prototype shall not be operated with a heated dipping bath or exposed to temperatures beyond the rated limits of its selected components. All acceptance testing will be performed using an unheated, laboratory-safe medium under room-temperature conditions.
- The prototype does not validate operation near molten wax, heated resin, process fumes, splashing hot material, or other industrial hot-dip conditions.

This limitation keeps the prototype safe and practical to demonstrate in the laboratory. Using molten wax or another heated material would introduce burn, splash, spill, and fume hazards that would require additional protective equipment and facilities. Heat could also affect the frame, wiring, sensors, motors, lubrication, and end effector.

Although these issues would need to be addressed in an industrial version, detailed thermal and mechanical design is outside the main scope of the ECET prototype. The project therefore focuses on demonstrating the electrical, control, sensing, communication, and safety systems at room temperature.

2.3 System Requirements

Each requirement below describes a specific, observable behavior with a defined pass condition. The system is treated as a black box for these requirements, and emphasis is placed on interactions with system boundaries, the required level of performance, and physical requirements.

Table 2.1
System requirements and verification criteria

ID	Requirement	Test
SR-01	The system shall not initiate motion unless a workpiece is detected at the input station.	Place a workpiece at the input station and command an operating cycle. The test passes if the system initiates the pick sequence. Remove the workpiece and command another cycle. The test passes if the system does not initiate motion.
SR-02	The system shall position the end effector at the programmed pickup position within ± 1 cm along both axes before acquiring the workpiece.	Command the system to the programmed pickup position and measure the position of the end effector relative to the target. The test passes if the measured error is within ± 1 cm along both axes.
SR-03	The system shall retain the workpiece throughout the complete pick-dip-place cycle without slipping or dropping it.	Execute one complete pick-dip-place cycle using the approved prototype payload. The test passes if the workpiece does not visibly slip or detach at any point during the cycle.
SR-04	The system shall lower the workpiece to the programmed immersion depth within ± 1 cm.	Execute one dip sequence and measure the actual immersion depth relative to the programmed value. The test passes if the measured depth is within ± 1 cm of the programmed depth.

ID	Requirement	Test
SR-05	The system shall hold the workpiece at the programmed immersion depth for the programmed dwell time within ± 1 s.	Execute one dip sequence and measure the dwell period with a stopwatch or equivalent timing instrument. The test passes if the measured dwell time is within ± 1 s of the programmed value.
SR-06	The system shall complete one full unattended pick-dip-place cycle, beginning with workpiece detection at the input station and ending with workpiece release at the output station and return to the home position.	Place a workpiece at the input station and command one operating cycle. Confirm that the system detects and acquires the workpiece, transfers it to the dipping position, lowers it to the programmed immersion depth, holds it for the programmed dwell time, raises it, transfers it to the output station, releases it, and returns to the home position. The test passes if every stage is completed in the correct order without operator intervention.
SR-07	The system shall remove power from both stepper-motor drivers when the emergency stop is actuated.	Actuate the emergency stop while an axis is moving and measure the power supplied to both stepper-motor drivers. The test passes if driver power is removed and all axis motion stops.

ID	Requirement	Test
SR-08	The system shall remain inhibited following an emergency-stop event until the emergency-stop button has been reset and the operator performs a deliberate restart action.	Actuate the emergency stop during operation, reset the emergency-stop button, and attempt to command motion without performing the restart action. The test passes if no motion occurs. Perform the deliberate restart action and confirm that normal operation can be re-established.
SR-09	The system shall transmit its current operating state and any active faults through the selected communications interface during operation.	Monitor the communications interface during one complete operating cycle and introduce at least one detectable fault condition. The test passes if each operating-state transition and the active fault are transmitted with the correct labels.
SR-10	The system shall complete ten consecutive unattended full pick-dip-place cycles using the approved prototype payload without dropping a workpiece, missing the intended placement position, or requiring operator intervention between cycles.	Command the system to perform ten consecutive full pick-dip-place cycles using the approved prototype payload. Record the cycle time, commanded immersion depth, measured immersion depth, and measured dwell time for each cycle. The test passes if all ten cycles are completed, no workpiece is dropped, no placement is missed, and the measured immersion depth and dwell time remain within their specified tolerances.

ID	Requirement	Test
SR-11	The system shall complete the full pick-dip-place cycle while carrying a workpiece with a mass of 1 lb.	Attach a 1 lb workpiece and command one complete pick-dip-place cycle. The test passes if the system lifts, transports, lowers, retains, and releases the workpiece without stalling, losing position, slipping, or dropping it.

2.4 Project Modules

The Safe Dip system can be divided into 7 modules as shown in Fig. 2.1.

The power distribution module is composed of the power supply and is in charge of converting wall power into the different types of voltage needed.

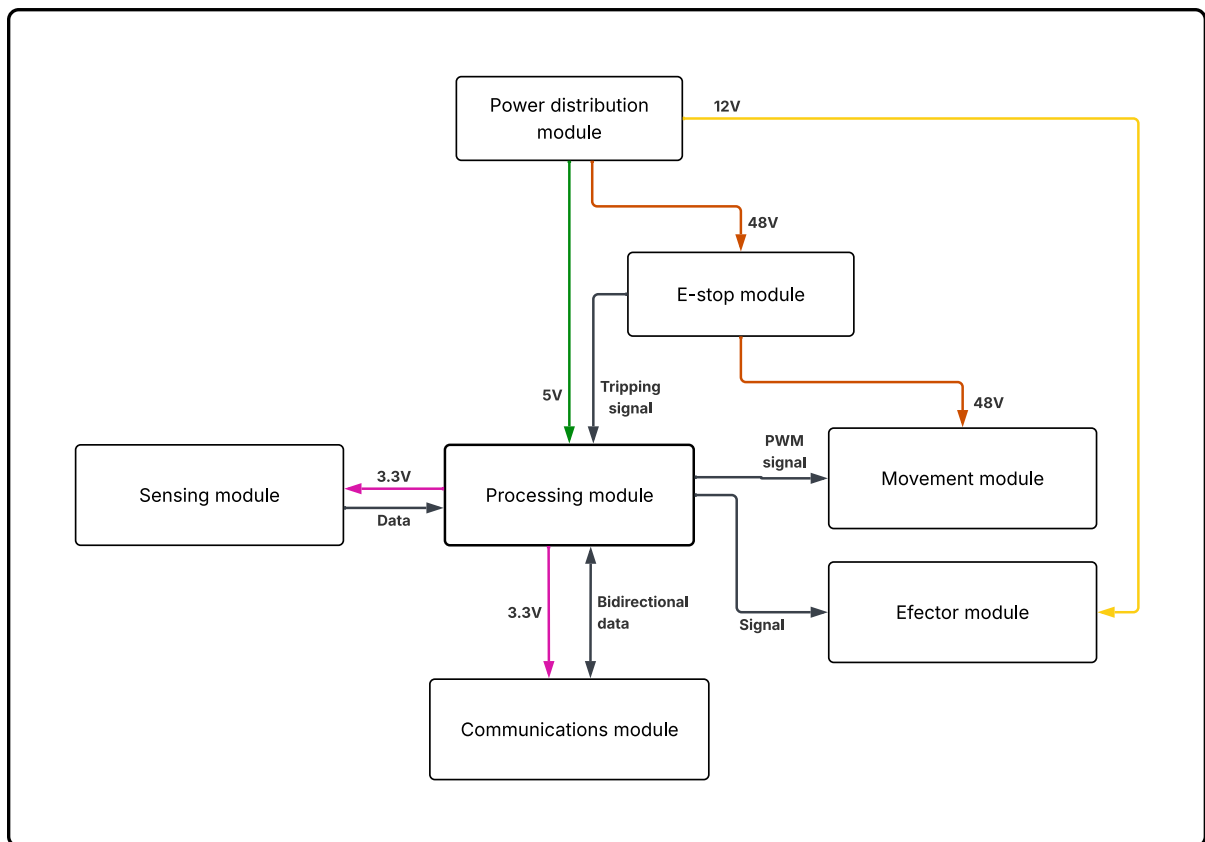


Fig. 2.1: System-wide block diagram

The E-stop module will contain all the mechanisms needed to perform a category-0 emergency stop, able to cut the power to the movement module. It is also in charge of alerting the processing module about the E-stop so that it can handle it gracefully.

The processing module is the heart of the system, encapsulating the microcontroller, which receives all inputs and translates them into the corresponding actuator responses. It also includes any and all components to ensure that the incoming data is at an acceptable input voltage for the microcontroller. This module receives its power from the power distribution module, and through the microcontroller's on-board power converter, it feeds 3.3 V to the other modules.

The sensing module is composed of all the time-of-flight (TOF) sensors used in this project, as well as limiting switches and other sensors.

The movement module is composed of all the stepper motors and controllers required for the horizontal movement of the gantry and the movement of the vertical assembly.

The effector module controls the part of the system that will grab the object and is composed of an electromagnet. It will, nonetheless, be designed to be as modular as possible, ensuring that a change of effector type may not be too hard to implement.

The communications module is an addition to the processing module in case communications are decided to be done via Ethernet TCP-Modbus. It would include a module capable of Ethernet communications.

The functional modules described above are expanded in Fig. 2.2 to show the proposed electrical implementation of Safe Dip and the interfaces between its major components. The diagram identifies the system's power-distribution paths, control signals, sensor connections, communication interfaces, and actuator drivers. It also distinguishes between the high-power circuitry used by the motion and end-effector subsystems and the low-voltage circuitry used by the processing, sensing, and communications modules. This hardware-level representation provides the basis for selecting components, defining subsystem specifications, preparing the power budget, and verifying that the modules can operate together as an integrated system.

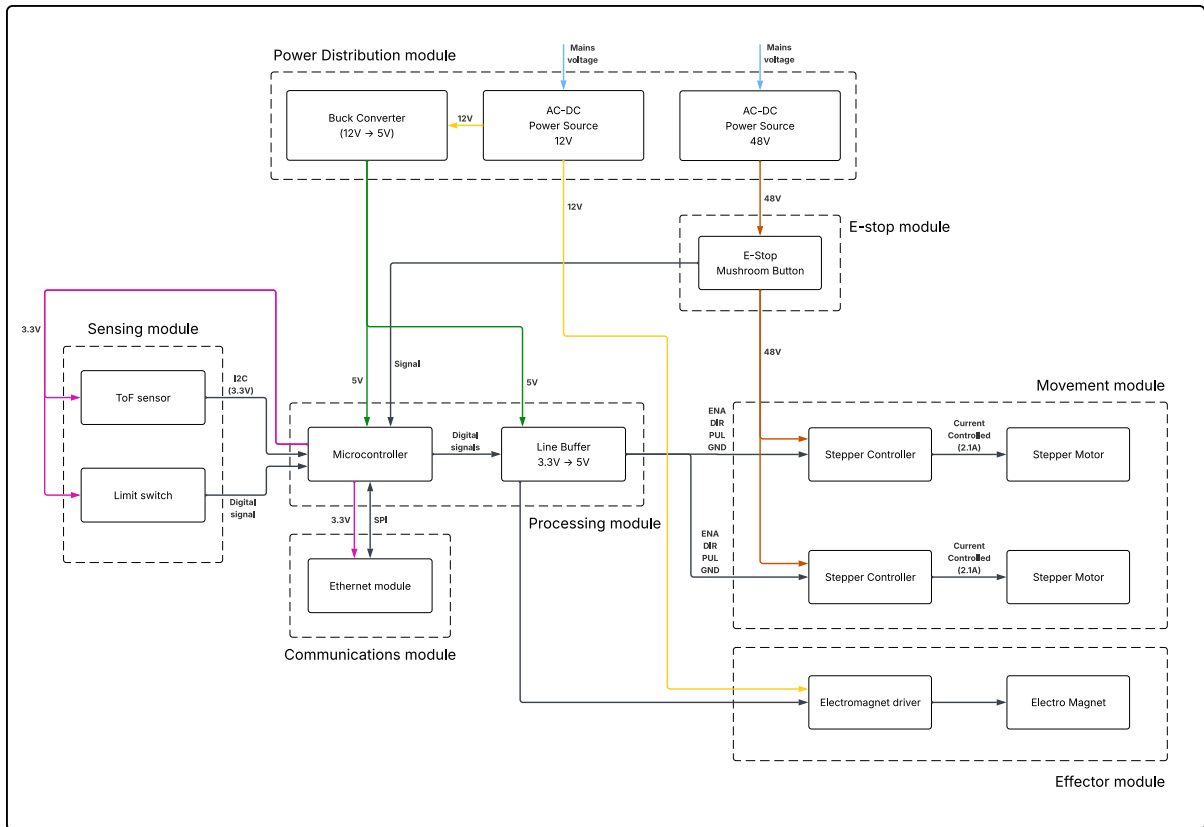


Fig. 2.2: Hardware architecture

2.5 System Specification

2.5.1 Power Distribution

Table 2.2
Power distribution specifications

Specific Component	Engineering Specifications	Justification & Verification
AC-DC power supply (motors)	• Input: 110 VAC, 60 Hz • Output: 24-48 VDC (preferentially 48 VDC) • Must supply sufficient current for two stepper motors at peak load	Verified by measuring output voltage under full load with a multimeter.
AC-DC power supply (12 V)	• Input: 110 VAC, 60 Hz • Output: 12 VDC • Ripple and noise within acceptable limits for logic, sensors, and effector	Verified by measuring output voltage and ripple with an oscilloscope under full load.
DC-DC Buck Converter (12 V $\rightarrow$ 5 V)	• Input: 12 VDC • Output: 5 VDC • Ripple and noise within acceptable limits for logic and sensors	Verified by measuring output voltage and ripple with an oscilloscope under full load.

2.5.2 E-Stop

Table 2.3
E-stop specifications

Specific Component	Engineering Specifications	Justification & Verification
Normally-closed emergency stop button	• Must be able to handle peak power coming from the AC-DC power supply (motors). • Requires manual reset before power is restored	Verified by actuating the E-stop during operation and confirming output power is cut.

2.5.3 Processing

Table 2.4
Processing specifications

Specific Component	Engineering Specifications	Justification & Verification
Microcontroller	• Operates at 3.3 VDC • Sufficient GPIO for stepper control, sensor inputs, and effector output • Supports USB UART for baseline serial communications • Capable of driving a Pulse Train Output (PTO) at the desired frequency set by the stepper and stepper controller	Verified by using an oscilloscope to check GPIO voltage, and an oscilloscope to verify PTO frequency.
Line buffer	• Input voltage: 3.3 VDC • Output voltage: 5 VDC	Verified by using a multimeter to measure the voltage on High and Low signals.

2.5.4 Sensing

Table 2.5
Sensing specifications

Specific Component	Engineering Specifications	Justification & Verification
Time of Flight (TOF) sensor	• Operating voltage: 3.3 VDC • Detection range sufficient to cover the input station • Blind zone under 3 cm • Communication protocol compatible with the microcontroller	Verified by placing and removing a part at the input station and confirming correct detection in both cases.
Limit switches	• Normally-closed configuration • One per axis end, used for homing • Rated for the operating environment	Verified by commanding each axis to its respective home and confirming motion stops upon switch actuation.

2.5.5 Movement

Table 2.6
Movement specifications

Specific Component	Engineering Specifications	Justification & Verification
Stepper motors (x2)	• NEMA 23 frame • Holding torque sufficient to lift rated payload on vertical axis and drive belt under load on horizontal axis • Positional accuracy: ± 1 cm	Verified by commanding each axis to a known target and measuring offset with a ruler.
Stepper motor drivers (x2)	• Compatible with 24-48 VDC supply • Microstepping support • Current output set to motor rating	Verified by confirming smooth motion on both axes across the full travel range.

2.5.6 End Effector

Table 2.7
End effector specifications

Specific Component	Engineering Specifications	Justification & Verification
Electromagnet	• Holding force sufficient to retain rated payload throughout the dip cycle • Controlled by a single digital output from the microcontroller	Verified by attaching a part at rated payload and confirming no drop occurs throughout one full pick-dip-place cycle.
MOSFET driver	• Logic-level gate compatible with the microcontroller's 3.3 V or 5 V output • Drain-source voltage rating greater than the 12 V electromagnet supply with adequate transient margin • Continuous drain-current rating greater than the maximum electromagnet current • Low on-state resistance at a gate voltage	Command the electromagnet on and off while monitoring the gate, drain, and electromagnet voltage with an oscilloscope. The test passes if the MOSFET switches the full electromagnet current reliably, remains within an acceptable temperature range, and no damaging voltage transient is observed when the electromagnet is de-energized.

2.5.7 Communications

Table 2.8
Communications specifications

Specific Component	Engineering Specifications	Justification & Verification
Ethernet module	• Operating voltage compatible with the processing module • SPI interface compatible with microcontroller • Reports system state, axis positions, and fault events • Supports Ethernet and TCP socket communication	Verify communication between the STM32 and Ethernet module through SPI. Establish a TCP connection from an external computer and confirm that data can be transmitted and received.

2.6 Power Budget

Table 2.9
Power budget for the Safe Dip prototype.

	Component	Voltage (V)	Current (A)	Unit Power (W)	Qty	Total Power (W)
Movement	Stepper motor (max usage per phase)	48	2.8	134.4	2	268.8
	Stepper motor driver	24–50 (48)	0.5	≈24	2	≈48
End Effector	Electromagnet	24	1.12	26.88	1	26.88
	Electromagnet driver	5	≈0.07–0.10	≈0.5	1	≈0.5
	Relays when powered	5	0.07–0.10	≈0.35–0.50	1	≈0.35–0.50
Sensing	Time-of-flight sensor	2.8–3.3	0.019	≈0.063	1	≈0.063
Communications	Ethernet module	3.3	0.132	≈0.436	1	≈0.436
Processing	STM32 (at max rate)	3.3/5	≈0.1	≈0.33	1	≈0.33
	Buffer ICs	5	≈2 μA	≈0.00001 (negligible)	1	≈0.00001 (negligible)
Total operational power						345.5 W

Table 2.10
Maximum power available from the Safe Dip AC-DC power adapters.

Power Rail	Power Adapter	Voltage (V)	Maximum Current (A)	Maximum Power (W)
48 V	Mean Well LRS-350-48	48	7.3	350.4
12 V	12 V AC-DC power adapter	12	5	60
Total available power				410.4 W

The Safe Dip prototype uses two independent AC-DC power adapters. The 48 V supply provides a maximum of 7.3 A, corresponding to approximately 350.4 W, and powers the movement subsystem. The 12 V adapter provides a maximum of 5 A, corresponding to 60 W, and powers the electromagnet and low-voltage circuitry. Together, the two adapters provide a total rated output capacity of approximately 410.4 W.

2.7 Project's Hardware

The components selected for Safe Dip were chosen to satisfy the engineering specifications defined in Section 2.5 while remaining within the project's budget constraint and accessible through standard commercial channels. The hardware is divided into two categories: the electronics responsible for control, motion, sensing, and communications, and the structural components that form the physical frame of the gantry.

2.7.1 Mechanical structure

The structural frame and motion transmission of Safe Dip are built from standard commercial extrusion and linear-motion components, selected so the preliminary dimensions listed in Table 2.11 can be met without custom machining.

The frame is assembled from 4040 T-slot aluminum extrusion, which forms the horizontal beam, the vertical column, and the supporting legs shown in Fig. 2.3. The T-slot profile allows components to be mounted at arbitrary positions along each beam and secured with T-nuts and M5 hardware, which simplifies alignment during assembly and permits adjustment of the input, dipping, and output station positions.

The horizontal axis is guided by an MGN12H linear rail and carriage and driven by a GT2 timing belt, converting the rotation of the horizontal NEMA 23 into the linear travel

between stations described in Table 2.11. The vertical axis uses a Tr8x2 leadscrew supported by pillow-block bearings and coupled to the vertical NEMA 23 through a flexible shaft coupler; the leadscrew converts motor rotation into the controlled lifting and immersion motion, and its selection is supported by the leadscrew speed and payload calculations in Appendix A. An anti-backlash nut block is used on the leadscrew to remove axial play so that programmed immersion depth remains repeatable across cycles.

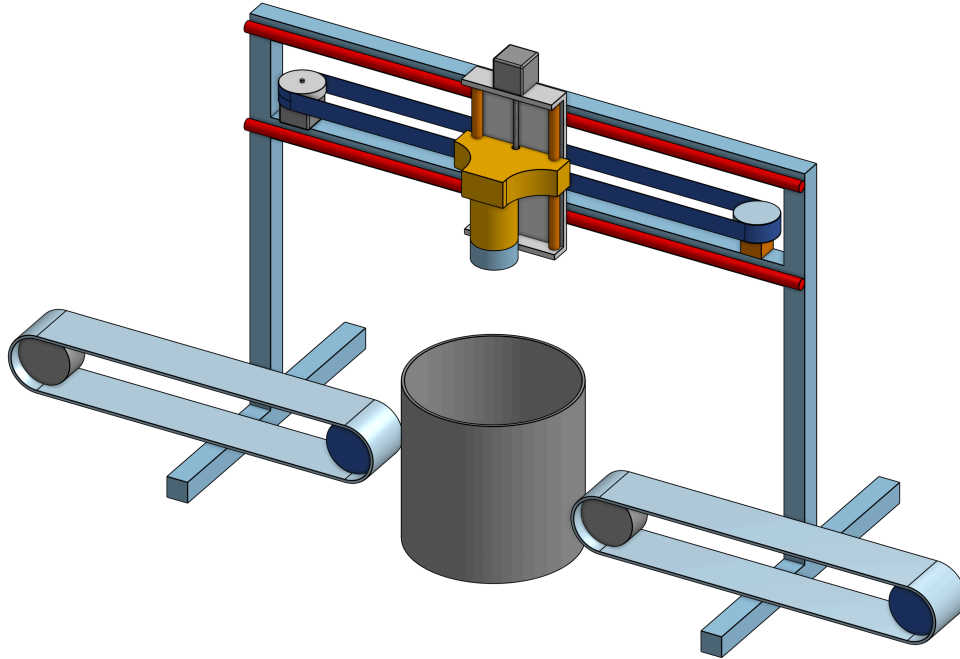


Fig. 2.3: Preliminary CAD model of the proposed Safe Dip mechanical arrangement -
Perspective view

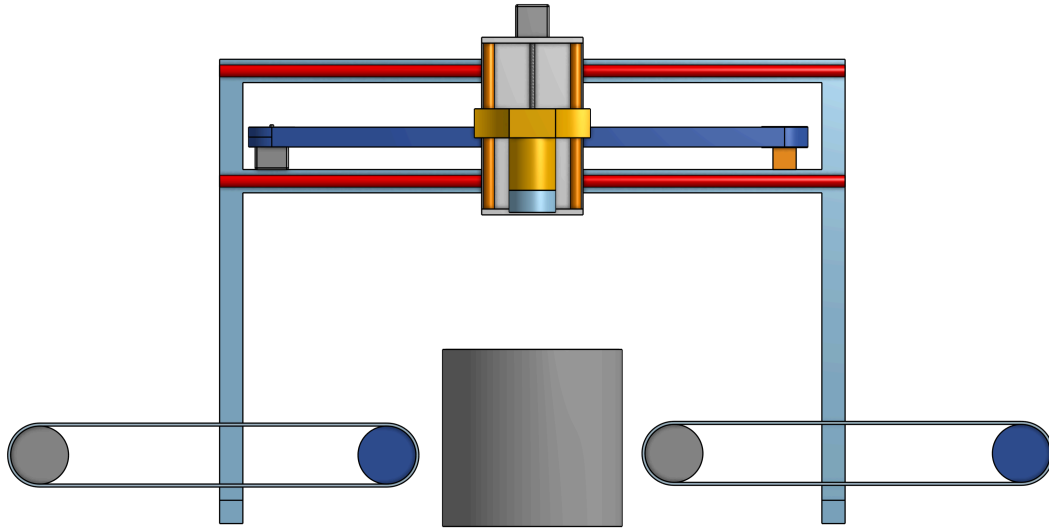


Fig. 2.4: Preliminary CAD model of the proposed Safe Dip mechanical arrangement -
Front view

The CAD model in Fig. 2.3 illustrates the proposed mechanical arrangement of the Safe Dip prototype. The model is preliminary and is intended to show the relative placement of the horizontal axis, vertical assembly, input station, dipping station, output station, and end effector. Final dimensions will be established after the selected mechanical components are measured and the required axis travel and clearances are verified.

The values in Table 2.11 are preliminary and are intended to describe the proposed operating envelope of the prototype rather than final fabrication dimensions.

Table 2.11
Principal preliminary dimensions of the proposed Safe Dip prototype.

Dimension	Preliminary Value	Design Purpose
Overall prototype dimensions	108 cm × 60 cm × 80 cm	Defines the approximate laboratory floor space and vertical clearance required by the prototype.
Usable horizontal travel	≈ 100 cm	Allows movement between the input, dipping, and output stations.
Usable vertical travel	≈ 30 cm	Provides sufficient lifting clearance and programmed immersion depth.
Bath envelope	30 cm × 30 cm × 30 cm	Defines the approximate size of the dipping container accommodated by the prototype.
Clearance above dipping bath	≈ 5 cm	Allows the workpiece and end effector to pass above the bath during horizontal transfer.
Maximum workpiece envelope	10 cm × 10 cm	Defines the approximate maximum workpiece size supported by the prototype geometry.

2.7.2 E-Stop and Protection module

The emergency-stop and protection hardware satisfies the requirements in Table 2.3. Its purpose is to remove power from the motor drivers independently of the STM32 while treating the electromagnet as a load-holding device. This arrangement stops the hazardous motion without causing the suspended workpiece to fall into the coating bath.

A normally closed, mechanically latching mushroom emergency-stop button is wired directly into the stepper-driver power path. When the button is actuated, its contacts open immediately and remove power from the DM542TE drivers, causing the axes to coast to a standstill and implementing a Category 0 stop. Because the stop path is hardwired and normally closed, either pressing the button or breaking a wire opens the circuit without requiring any action from the microcontroller. The button remains latched until it is manually reset, satisfying the “requires manual reset” requirement in Table 2.3.

This stop path is intentionally independent of firmware and electronic control logic. IEC 60204-1 requires a Category 0 emergency stop to operate through hardwired electro-mechanical components rather than depending on electronic logic [4]:.

A separate relay controls the electromagnet coil supply and is not included in the Category 0 cutoff. The electromagnet is treated as a load-holding device rather than as a hazardous actuator that must be immediately de-energized. Removing power from it during an emergency stop could create a secondary hazard by releasing the suspended ferromagnetic part into the coating bath. IEC 60204-1 requires the emergency-stop function to avoid creating such additional hazards [4]. The electromagnet therefore remains energized during the stop, and the part is released in a controlled manner only after motion has ceased.

An inline fuse holder provides overcurrent protection for the switched supply rail, while the resistor kit provides the gate and pull resistors required by the relay and associated switching stages. A passive buzzer provides an audible indication whenever a fault or emergency-stop condition is active.



Fig. 2.5: E-stop mushroom button

2.7.3 Sensing Module

The sensing hardware provides the workpiece-presence and axis-homing feedback required for the sense-pick-dip-place sequence and satisfies the requirements in Table 2.5.

A VL53L0X time-of-flight sensor detects the presence of a workpiece at the input station before a cycle begins. It operates at 3.3 V and communicates over I2C, making it directly compatible with the STM32 without additional level shifting. Its short blind zone and detection range are suitable for covering the input station as required in Table 2.5.

Normally-closed KW12-3 micro limit switches provide the homing and travel references for each axis. One switch is placed at an axis end so that, on startup, each axis can be driven toward it to establish a known home position; the normally-closed configuration ensures that a disconnected or failed switch is detected as an actuated state rather than being ignored. Together with the ± 1 cm positional accuracy of the movement subsystem (Table 2.6),

this homing reference allows the controller to reach the input, dipping, and output stations repeatably.



Fig. 2.6: vl53l0x ToF sensor

2.7.4 Power Distribution Module

This system will require three main voltage lines, one at 24-48 V for the stepper motor controllers, another at 12 V for the effector, and finally one of 5 V for the processing module. To that end, two different power supplies will be used for the 48 V and 12 V lines, while a DC-DC buck converter will be responsible for generating the 5 V line.

The Mean Well LRS-350-48 is an AC-DC power supply that converts 110 or 220 Vrms into 48 V. It has four parallel terminals for the output, and it can supply a maximum current of 7.3 A.



Fig. 2.7: Mean Well LRS-350-48 AC-DC power supply

The less stringent requirements of the 12 V line means that it can be provided by an off-brand transformer, as long as it complies with the specifications. In this case, an ALITOVE DC 12V 5A Power Supply was chosen.



Fig. 2.8: ALITOVE DC 12V 5A Power Supply

Finally, for the 5 V line, as shown in Fig. 2.2, a HiLetgo buck converter based on the XL4016 chip was chosen.

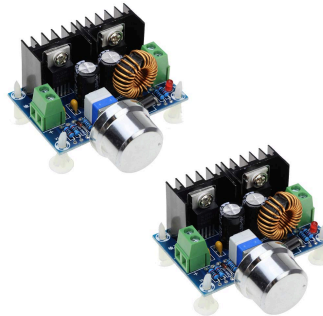


Fig. 2.9: HiLetgo buck converter

2.7.5 Processing Module

The STM32 NUCLEO-F446RE serves as the microcontroller responsible for coordinating the efforts for the sense-pick-dip-place sequence. It features an ARM Cortex-M4, 32-bit, 3.3 V microcontroller operating at a maximum frequency of 180 MHz. The STM32 family is widely adopted in embedded industrial designs due to its peripheral density, which allows for tight deterministic loops.

This board provides the necessary computing bandwidth to manage the real-time state machine while simultaneously running communication tasks. Crucially, its advanced hardware timers can be utilized in Output Compare mode to generate precise, jitter-free frequency pulses for step generation without taxing the CPU. Furthermore, its configurable

interrupts can be used to immediately respond to external signals or to timer completion. The ability to delegate tasks to specialized hardware enables the Nucleo board to perform communication tasks with other peripherals.

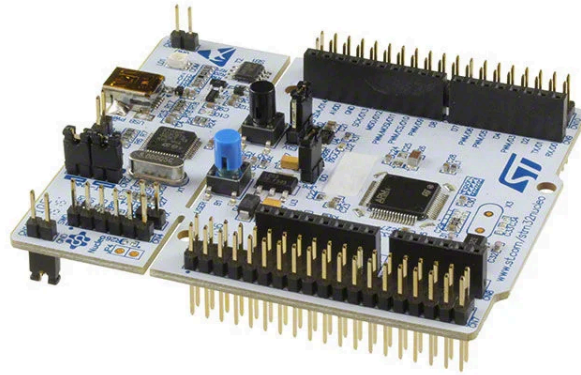


Fig. 2.10: STM32 Nucleo-F446RE microcontroller board

Given that the Nucleo board operates using 3.3 V logic, it will require a buffer for communicating with the stepper controllers. To that end, something like the 74HCT244 for boosting the signal to 5 V.

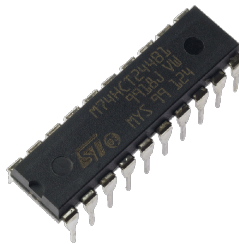


Fig. 2.11: 74HCT244 Octal Buffers and Line Drivers

2.7.6 Movement Module

For both the horizontal and vertical movement, a NEMA 23 stepper motor will be used for each axis. For this project, StepperOnline's 23HS22-2804S NEMA 23 stepper motor was chosen due to its static torque of 1.24 Nm and dynamic torque of almost 0.9 Nm at 150 RPM.



Fig. 2.12: NEMA 23 23HS22-2804S stepper motor

To drive the NEMAs using a microcontroller, a stepper motor controller is required to set their speed and direction. The DM542TE is capable of tolerating up to 48 V and accepting 5 V signals.



Fig. 2.13: DM542TE stepper controller

2.7.7 End Effector Module

The end-effector hardware holds the ferromagnetic workpiece throughout the pick-dip-place cycle and switches it under microcontroller control, satisfying the requirements in Table 2.7.

A Baomain BM-P 80-38 electromagnet is used as the gripping element. It holds the ferromagnetic workpiece by magnetic attraction rather than a mechanical grip, which avoids moving jaws near the coating bath and lets the part be released simply by removing coil current. Its holding force is sized to retain the rated payload for the full duration of the dip, as required in Table 2.7, and it is driven from a single microcontroller output.

An IRLZ44N logic-level MOSFET switches the electromagnet coil from a single 3.3 V/5 V STM32 GPIO. It was chosen because its logic-level gate is compatible with the microcontroller output, and its drain-source voltage and continuous drain-current ratings comfortably exceed the 12 V electromagnet supply and coil current with transient margin, matching the MOSFET-driver requirements in Table 2.7.

A 1N4007 flyback diode is placed across the electromagnet coil to protect the MOSFET and GPIO from the inductive back-EMF generated when the coil is de-energized, addressing the “no damaging voltage transient” condition in the Table 2.7 verification.



Fig. 2.14: Baomain BM-P 80-38 electromagnet

2.7.8 Communications Module

Communications between Safe Dip and the user are divided into two categories for this project: a base case and an extension goal. For the base case, the NUCLEO-F446RE's UART peripheral, along with the ST-LINK connector on the board, will be used.

For the extension goal to communicate using Modbus-TCP, a W5500 was chosen. The W5500 is a standalone Ethernet module with a full TCP/IP stack implemented in the hardware/firmware on the chip itself. This means that the microcontroller doesn't have to run a TCP/IP stack, but instead it only has to communicate with the module via SPI.

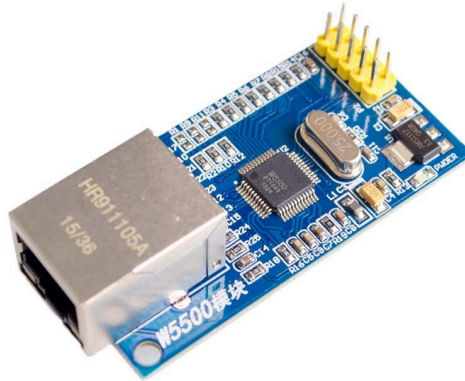


Fig. 2.15: W5500 Ethernet module

2.8 Project's Software

2.8.1 Software Platform, Libraries, and Integration

The Safe Dip firmware will be developed for the STM32 NUCLEO-F446RE using STM32CubeMX and the STM32 Hardware Abstraction Layer (HAL). STM32CubeMX will be used to configure the microcontroller clock, GPIO pins, hardware timers, external interrupts, UART, and SPI peripherals. It will also generate the initial peripheral setup code, reducing

the amount of low-level configuration that must be written manually and helping ensure that the selected pins and peripheral functions are compatible.

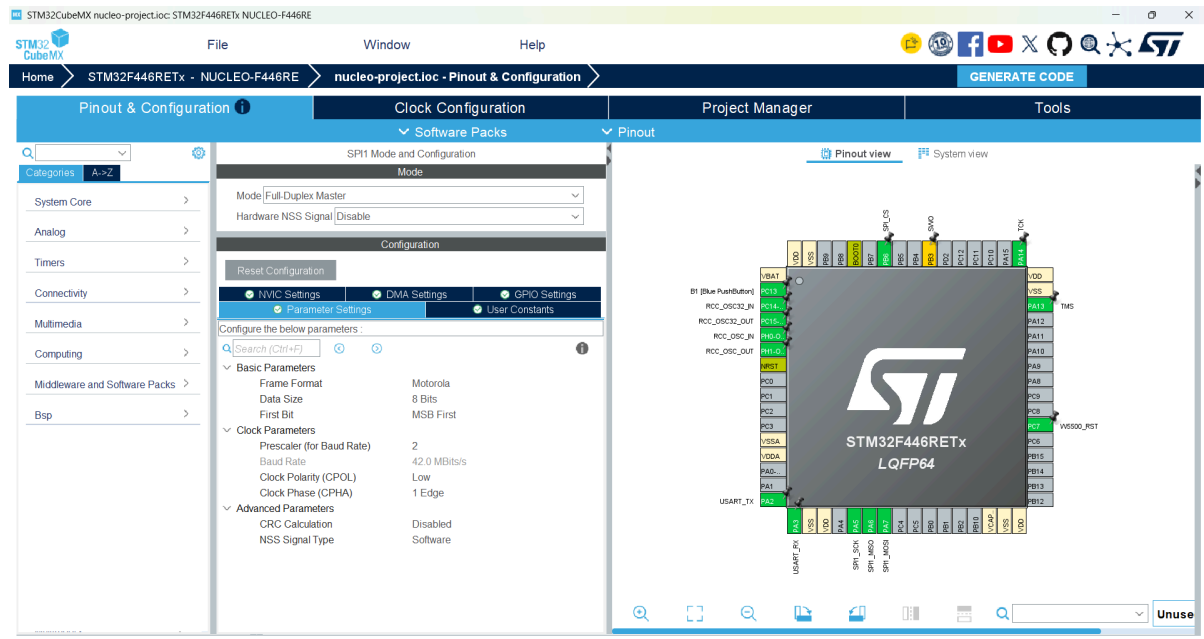


Fig. 2.16: STM32CubeMX graphical interface

The STM32 HAL library will provide the main interface between the application software and the microcontroller peripherals. HAL functions will be used to initialize and control the timers responsible for generating the stepper-motor STEP signals, configure the direction and end-effector outputs, read sensor and limit-switch inputs, and communicate with external devices. Interrupt callbacks provided by HAL will also be used for events such as emergency-stop activation, timer completion, and communication activity. Application-specific behavior, including movement sequencing and safety handling, will remain separate from the generated HAL initialization code.

Ethernet communication will be implemented using the WIZnet W5500 controller. The W5500 communicates with the STM32 through SPI and provides hardware support for Ethernet, IP, TCP, and UDP communication. The WIZnet socket library will be used to configure the controller and manage its hardware sockets without requiring the STM32 to execute a complete software TCP/IP stack.

A small integration layer will connect the WIZnet library to the STM32 HAL implementation. This layer will provide the SPI transfer operations, chip-select control, controller reset, timing functions, and any interrupt handling required by the W5500 library. Keeping this code separate from both the vendor library and the application logic will make it easier to update or replace either side without modifying the complete communications implementation.

Modbus TCP will be implemented above the W5500 socket interface. The communications module will translate between Modbus requests and the internal state of the Safe Dip system, exposing information such as the operating mode, active instruction, gantry position, sensor values, and fault status. Commands received through Modbus will be validated before being passed to the control software so that communications code does not directly manipulate motors, timers, or other hardware peripherals.

Together, STM32CubeMX, the STM32 HAL, the WIZnet library, and the custom integration code provide a layered software platform. Vendor libraries will handle hardware-specific peripheral access, while the project-specific modules will implement movement control, end-effector control, safety behavior, sequencing, and communications.

2.8.2 Software Architecture and Control Flow

The Safe Dip control software will be implemented on the STM32 microcontroller using a cooperative, nonblocking architecture. Rather than executing the complete pick-dip-place cycle as one blocking function, the software will be divided into several modules that are processed repeatedly from the main loop. These modules include the cycle sequencer, movement controller, end-effector controller, dwell timer, and communications interface.

Each processing function will perform a bounded amount of work and return immediately. This allows the controller to remain responsive to safety inputs and communication requests while a movement, effector operation, or timed operation is still in progress.

As shown in Fig. 2.17, the software first initializes the microcontroller peripherals and the internal state of each module. It then enters a continuous main loop. During each iteration, the program steps the cycle sequencer, processes the movement module, processes the end-effector module, updates the dwell timer, and services the communications interface.

This structure implements cooperative multitasking. Each module is given an opportunity to perform its current work, but no module is allowed to block execution while waiting for an operation to finish. The movement, effector, and timer modules therefore continue operating over multiple iterations of the main loop while the sequencer monitors the progress of the current instruction.

The movement module will be responsible for the low-level control of the two stepper motors. Its responsibilities include setting the direction signals, generating the STEP waveforms using STM32 hardware timers, counting completed steps, applying acceleration and deceleration profiles, and stopping each axis at the requested position. The end-effector module will control the engagement and disengagement of the magnetic tool, while the timer module will manage the programmed dwell period.

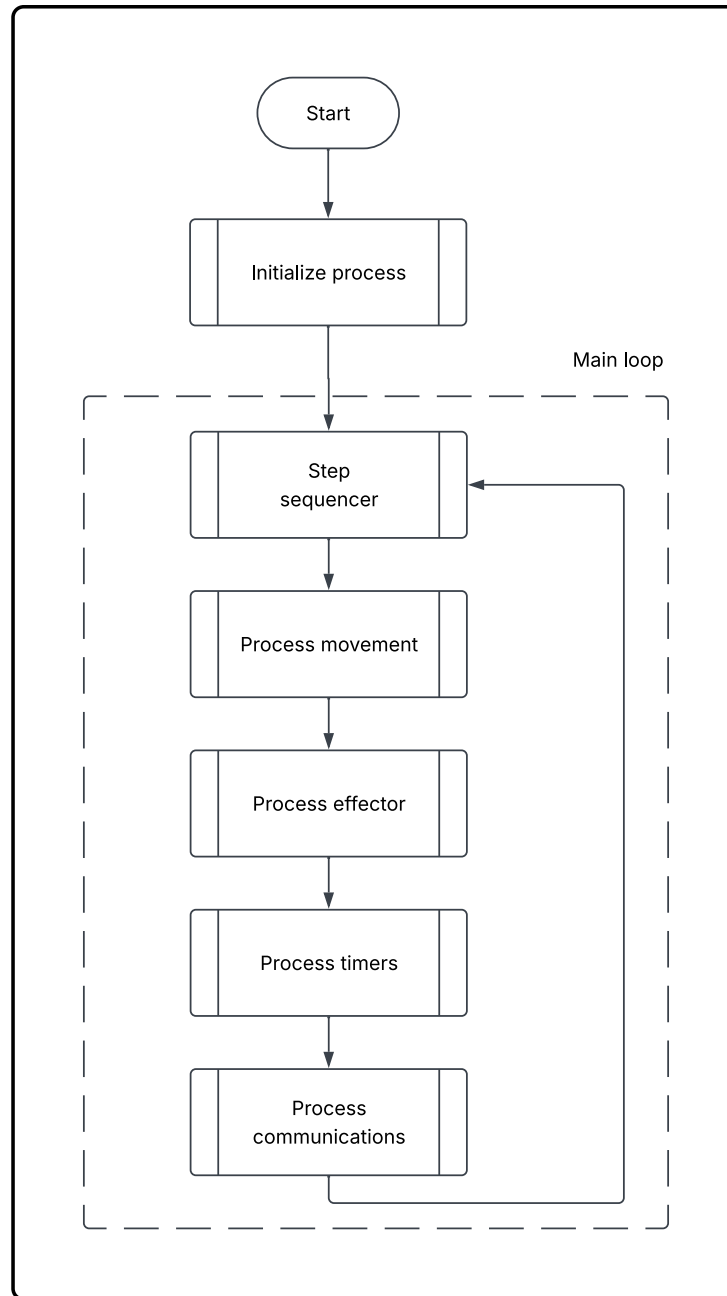


Fig. 2.17: Main software execution structure

These implementation details will remain encapsulated inside their respective modules. The cycle sequencer will not directly manipulate timer registers, GPIO pins, motor step counts, or acceleration parameters.

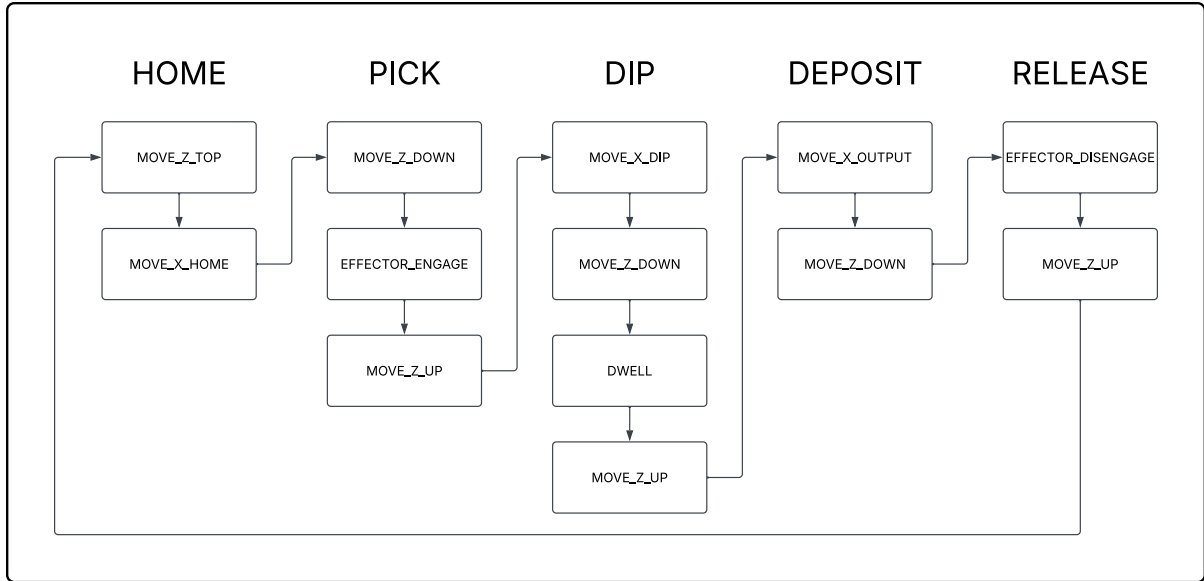


Fig. 2.18: Normal pick-dip-place instruction sequence

2.8.2.1 Normal Cycle Sequence

The normal operating cycle is represented as an ordered sequence of nonblocking instructions, as shown in Fig. 2.18. The instructions are grouped into five process stages: home, pick, dip, deposit, and release.

Each instruction represents an absolute objective rather than a blocking procedure. For example, `MOVE_Z_UP` requests that the vertical axis move to its configured upper position. The movement module starts the operation and returns immediately. The sequencer continues to monitor the instruction until the movement module reports that the target has been reached.

2.8.2.2 Cycle Sequencer

The cycle sequencer maintains an instruction index identifying the current entry in the normal sequence. It also evaluates the state of the current instruction. An instruction may be `IDLE`, `RUNNING`, `FINISHED`, or `ABORTED`, and it is the corresponding module's job to indicate what its state is, as later shown in Fig. 2.21.

While the instruction is running, the sequencer processes its current status but does not start another instruction. In this condition, the sequencer effectively performs no additional sequencing work during that iteration. Control immediately returns to the main loop so that movement, effector, timer, and communication processing can continue.

When the responsible module reports that the instruction has finished, the sequencer advances the instruction index and starts the next entry in the normal sequence. Once the final instruction has been completed, the index returns to the beginning of the sequence.

If the system enters the stopping condition, the sequencer sends abort requests to the movement, effector, and timer modules. Once the system has stopped, the current instruction is considered aborted, and it is the module's responsibility to report it as such. The instruction index is reset before another normal cycle is permitted to begin. This prevents the system from continuing from the middle of an instruction whose physical state may no longer be known.

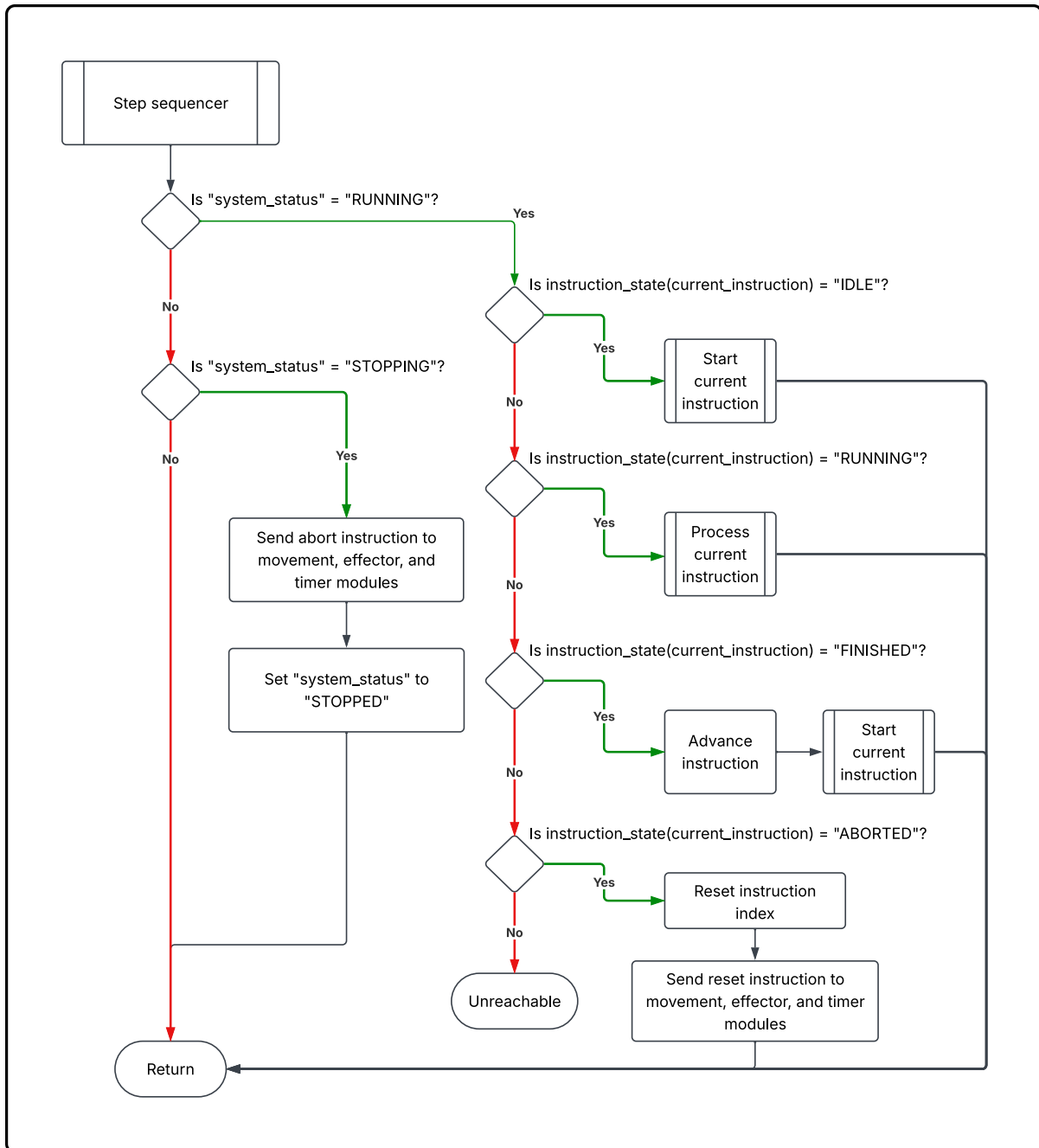


Fig. 2.19: Cooperative cycle sequencer

2.8.2.3 Starting an Instruction

Instruction dispatch is separated from the internal implementation of the movement, end-effector, and timer modules. The sequencer identifies which module is responsible for the current instruction and communicates with that module only through its public interface.

As illustrated in Fig. 2.20, instructions beginning with `MOVE_X_*` or `MOVE_Z_*` are dispatched to the movement module. Instructions beginning with `EFFECTOR_` are dispatched

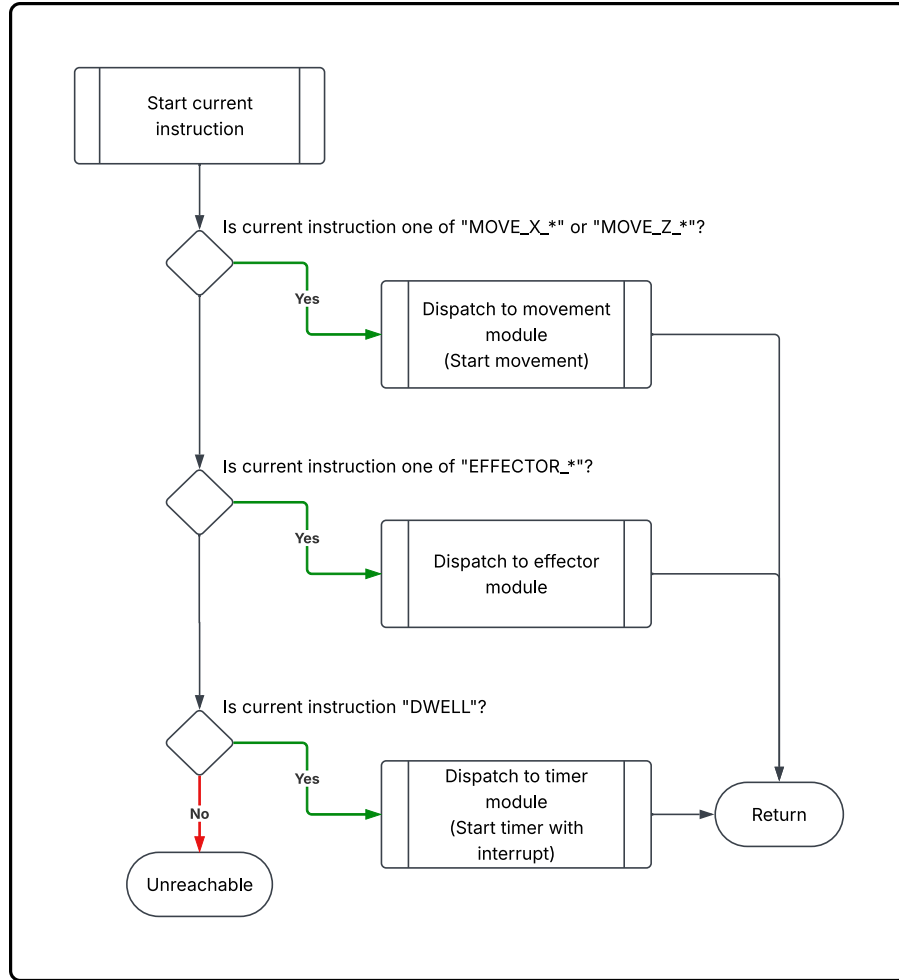


Fig. 2.20: Dispatching the current instruction to its responsible module

to the end-effector module. The DWELL instruction is dispatched to the timer module. Each module provides a start operation that configures the requested action and returns immediately. For example, starting a movement may configure the required direction, target step count, acceleration profile, and timer channel before enabling the STEP waveform. The function does not wait for the motor to reach its target.

Similarly, starting the dwell instruction configures and starts the timer without waiting for the timer to expire. Consequently, dispatching an instruction does not block execution of the cooperative main loop.

2.8.2.4 Instruction-State Evaluation

Once an instruction has been started, the sequencer must determine whether its responsible module is still running, has completed, or has been aborted.

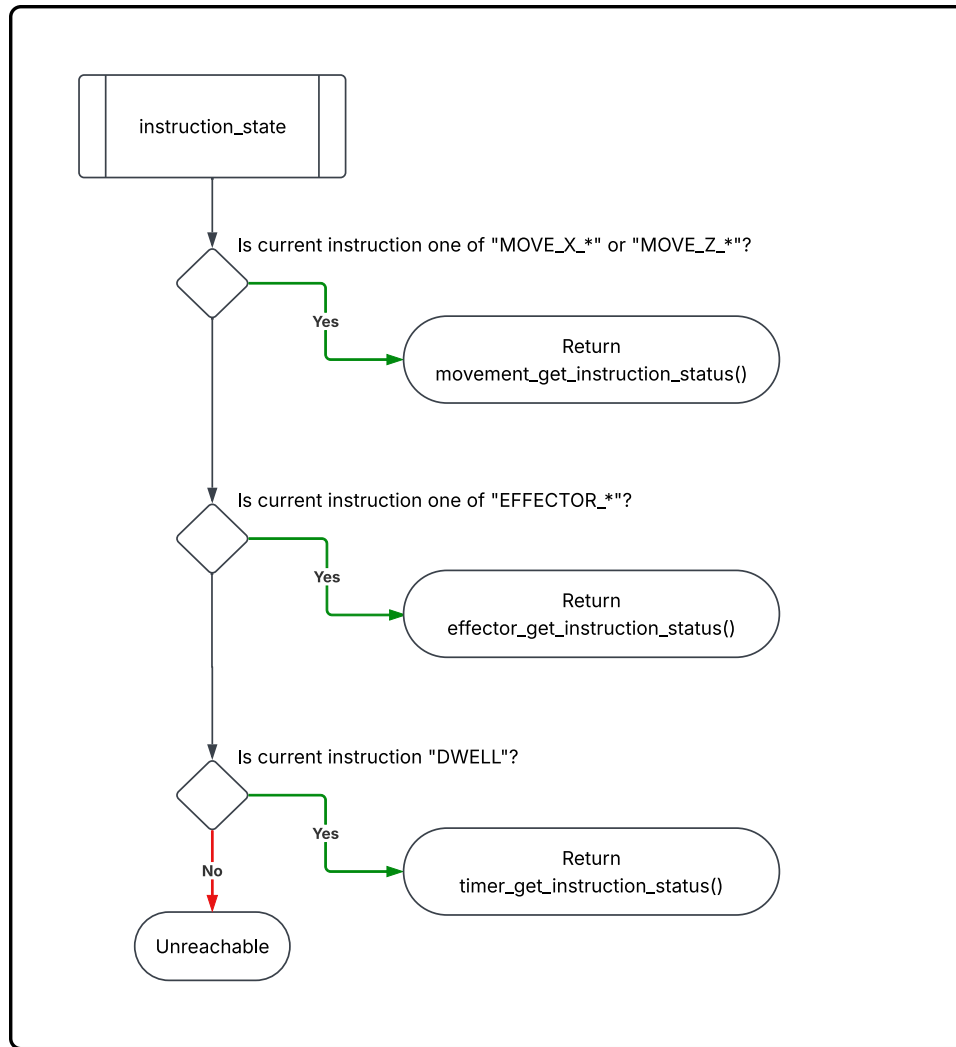


Fig. 2.21: Evaluation of the current instruction state

The instruction-state function uses the same instruction classification as the dispatch operation. For a movement instruction, it returns the status reported by the movement module. For an end-effector instruction, it returns the status reported by the end-effector module. For the DWELL instruction, it returns the status reported by the timer module.

The sequencer therefore coordinates the complete production cycle without knowing how the individual modules implement their operations. It does not need to know which hardware timer is assigned to an axis, which GPIO pins control the stepper drivers, how many steps remain, or how the acceleration profile is calculated.

This produces a clear abstraction boundary. The sequencer knows the public interfaces provided by the movement, effector, and timer modules, but those modules do not need to know the structure of the complete pick-dip-place sequence. The modules can therefore

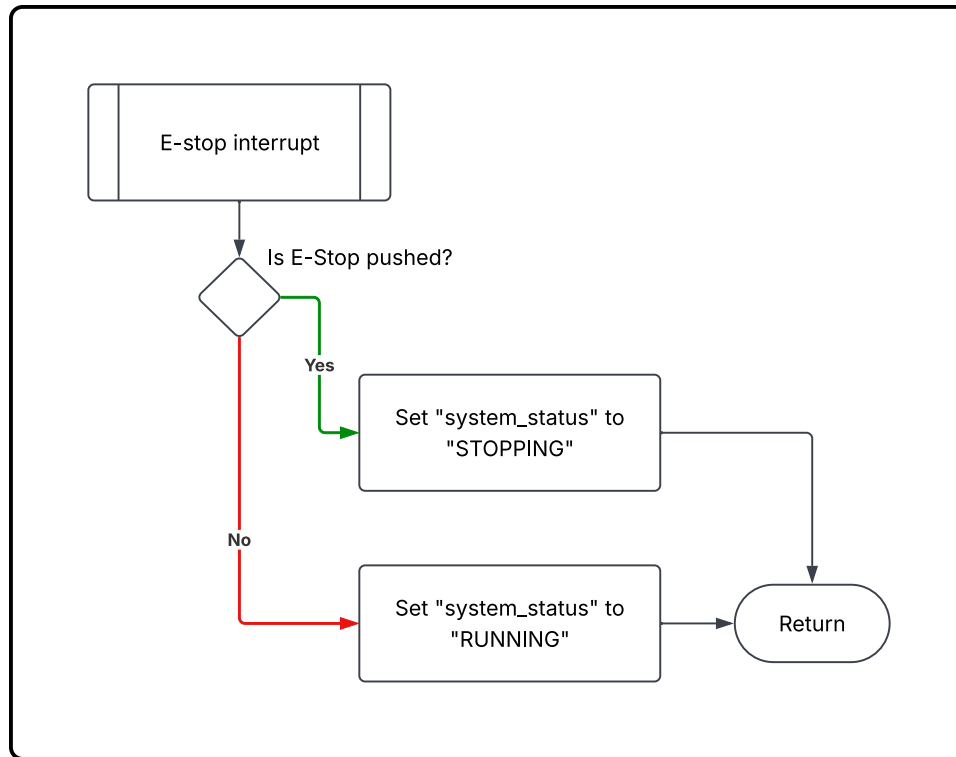


Fig. 2.22: Emergency-stop interrupt handling

be developed and tested independently by issuing commands and observing their reported status.

2.8.2.5 Emergency-Stop Handling

The emergency-stop input is processed asynchronously using an external interrupt. This allows the microcontroller to detect the E-stop regardless of which instruction or main-loop function is currently being processed.

The interrupt service routine performs only the minimum processing required to record the state of the E-stop input. When the E-stop is asserted, the routine sets the system status to STOPPING and returns. It does not directly execute the complete stopping sequence inside the interrupt context.

During the next execution of the sequencer, the STOPPING status causes abort requests to be sent to the movement, end-effector, and timer modules. Each module is responsible for placing its own operation and peripherals into the requested stopped or safe condition. The movement module, for example, will stop the active acceleration profile and disable the appropriate STEP-generation timers.

After the abort requests have been processed, the software changes the system status to STOPPED. The current sequence instruction is then marked as aborted, and the instruction index is reset before another normal cycle can begin.

Releasing the E-stop will not automatically resume the interrupted instruction. A deliberate operator reset will be required before operation is restored. The software will then re-establish a known gantry position and restart the sequence from a defined boundary rather than attempting to continue from a partially completed movement.

The software-controlled response supplements the independent hardware safety circuitry. The hardwired E-stop mechanism remains responsible for enforcing the final removal of actuator power if the controlled software response cannot be completed within the permitted time.

Overall, this architecture combines cooperative scheduling with asynchronous peripheral operation. Hardware timers and interrupts perform timing-sensitive work, while the main loop coordinates higher-level operations through the sequencer. This approach keeps the normal production sequence easy to understand while isolating the detailed control of the motors, end effector, dwell timer, and communications interface inside dedicated software modules.

2.9 Preliminary work

2.9.1 Modbus TCP Communications Prototype

Preliminary work has been completed on the communications subsystem through the development of a portable Modbus server library in C. The library separates the Modbus protocol from the target hardware by receiving raw protocol data units and accessing application data through user-provided callback functions. This allows the same protocol implementation to be tested on a desktop computer and later integrated into the STM32 firmware without introducing hardware-specific dependencies into the library. The implementation avoids dynamic memory allocation and organizes the supported Modbus function codes into separate modules. :contentReference[oaicite:0]{index=0}

The library currently includes dispatch support for coils, discrete inputs, input registers, and holding registers. Incoming requests are validated before application data is accessed, and protocol exception responses are generated for unsupported functions, invalid addresses, invalid values, and internal failures. The public interface exposes a central PDU dispatcher, while a context structure containing read and write callbacks connects Modbus addresses to the Safe Dip application state. :contentReference[oaicite:1]{index=1} :contentReference[oaicite:2]{index=2} :contentReference[oaicite:3]{index=3}

A Modbus TCP layer has also been implemented to process the Modbus Application Protocol header and forward the enclosed PDU to the protocol dispatcher. This layer uses a small state machine that first receives and parses the seven-byte MBAP header and then requests the remaining PDU bytes. This structure supports nonblocking communication because the STM32 does not need to wait for an entire request in one operation. :contentReference[oaicite:4]{index=4} :contentReference[oaicite:5]{index=5}

The TCP implementation was integrated with the STM32 and W5500 Ethernet library through SPI. In this arrangement, the W5500 hardware manages the Ethernet and TCP socket operations, while the STM32 periodically checks the socket state, receives available request bytes, dispatches completed Modbus messages, and transmits the resulting responses without blocking the main control loop.

Initial hardware testing was performed using `madbus-cli` as a Modbus TCP client. The test successfully read coil values, wrote individual coils, wrote several coils in one request, and read the values back from the STM32. The results shown in Fig. 2.23 confirm preliminary operation of the read-coils, write-single-coil, and write-multiple-coils functions through the complete computer-to-W5500-to-STM32 communication path.

```
> madbus-cli --ip 192.168.0.2 read-coil 0 4
Coil states from unit 1:
    10000: false
    10001: false
    10002: false
    10003: false
> madbus-cli --ip 192.168.0.2 write-coil 0 true
Unit 1 wrote the coil 10000 to true
> madbus-cli --ip 192.168.0.2 read-coil 0 4
Coil states from unit 1:
    10000: true
    10001: false
    10002: false
    10003: false
> madbus-cli --ip 192.168.0.2 write-coil 1 true true
Unit 1 wrote the registers 40001..40001
> madbus-cli --ip 192.168.0.2 read-coil 0 4
Coil states from unit 1:
    10000: true
    10001: true
    10002: true
    10003: false
> madbus-cli --ip 192.168.0.2 write-coil 0 true false true true
Unit 1 wrote the registers 40000..40003
> madbus-cli --ip 192.168.0.2 read-coil 0 4
Coil states from unit 1:
    10000: true
    10001: false
    10002: true
    10003: true
```

Fig. 2.23: Preliminary Modbus TCP coil read and write testing using `madbus-cli`

This preliminary implementation reduces the remaining communications work to defining the final Modbus address map and connecting its callbacks to the Safe Dip operating state, gantry position, sensor information, configuration values, and fault data.

2.9.2 Stepper-Motor Signal Prototype

Preliminary work was also performed on the generation of the stepper-motor STEP signal using the STM32 hardware timers. TIM2 was configured in Output Compare toggle mode so that the timer peripheral, rather than the main application loop, controls the output transitions. After each compare event, the interrupt callback schedules the next transition by advancing the compare register by a configurable number of timer ticks. Changing this interval changes the STEP frequency without requiring software delays or a blocking pulse-generation loop.

This experiment establishes the basic mechanism that will later be incorporated into the movement module. The module will extend the prototype with pulse counting, acceleration and deceleration, target-position tracking, and independent control of both axes. Testing performed at this stage was limited to timer configuration and STEP waveform generation; operation with the selected stepper drivers and motors remains part of the final implementation and verification.

```

1 // Dynamically controls the half period
2 static volatile uint32_t half_period_ticks = 500U;
3
4 // Stepper start function
5 void stepper_start(void) {
6     // Get counter
7     uint32_t now = __HAL_TIM_GET_COUNTER(&htim2);
8
9     // Set Capture/Compare Register as `now + half_period_ticks`
10    __HAL_TIM_SET_COMPARE(&htim2, TIM_CHANNEL_1, now + half_period_ticks);
11
12    // Start timer + interrupt
13    HAL_TIM_OC_Start_IT(&htim2, TIM_CHANNEL_1);
14 }
15
16 // Timer compare interrupt
17 void HAL_TIM_OC_DelayElapsedCallback(TIM_HandleTypeDef *htim) {
18     // Check it comes from correct timer and channel
19     if ((htim->Instance == TIM2) && (htim->Channel ==
20         HAL_TIM_ACTIVE_CHANNEL_1)) {
21         // Get Capture/Compare Register
22         uint32_t previous_compare = __HAL_TIM_GET_COMPARE(htim,
23             TIM_CHANNEL_1);
24         // Set updated Capture/Compare Register
25         __HAL_TIM_SET_COMPARE(htim, TIM_CHANNEL_1,
26             previous_compare + half_period_ticks);
27     }
28 }

```

Fig. 2.24: Snippet for STEP square wave generator

Chapter 3

Non-Technical Considerations

Summary

This chapter presents the project planning and non-technical considerations associated with Safe Dip. It provides a preliminary monetary budget, a sixteen-week implementation timeline, and the proposed distribution of responsibilities between the team members. The chapter then evaluates the environmental, health and safety, ethical, social, and sustainability aspects of the system. Particular attention is given to the hazards created by moving machinery, electrical power, and the handling of an approximately 10 kg suspended load for the fully-featured system, along with the prototype controls used to reduce those risks. Finally, the chapter discusses the project's potential to reduce worker exposure to hazardous and repetitive dipping operations while emphasizing honest reporting, maintainability, and responsible automation.

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3.1 Monetary Budget

The monetary budget in this section represents a preliminary estimate for the laboratory prototype. Prices were obtained from currently available commercial listings and may change as component availability, shipping costs, taxes, and final mechanical dimensions are resolved. The external conveyor is treated as supporting equipment rather than part of Safe Dip and is therefore calculated separately from the prototype total.

Table 3.1
Project component budget

Item	Component	Qty.	Price total
Single Output Switchable Power	Mean Well LRS-350-48	1	\$35.06
Microcontroller	STM32 NUCLEO-F446RE	1	\$31.00
4-channel digital buffer/line driver	SN74AHCT125N Quadruple Buffer	1	\$0.80
Stepper Motor	3HS22-2804S NEMA 23	2	\$26.74
Stepper Controller	DM542TE Stepper Controller	2	\$42.00
Ethernet Module	W5500 Ethernet LAN Module	1	\$6.50
E-Stop	1NC 1NO Red Mushroom Emergency Stop	1	\$9.99
Transformer	ALITOVE Power Supply Adapter Converter Transformer	1	\$14.00
Buck Converter	LM2596 DC-DC Buck Converter	1	\$1.60
Electromagnet	Electromagnet Solenoid BM-P 80-38	1	\$44.98
Proximity Laser Sensor	VL53L0X ToF Sensor	1	\$2.98
Shaft Coupler	Shaft Coupler (6.35 x 8 mm)	2	\$11.99
Limit Switch	HiLetgo KW12-3 Micro Limit Switch	1	\$0.60
Rail Guide	MGN12H 300 mm Linear Rail Guide	2	\$35.99
Lead Screw	TR8x2 Lead Screw (300 mm)	1	\$11.99
Bearings	Pillow Block Bearing, 5/16 in (8 mm)	4	\$8.49
Guide Rails	Linear Rail Slide SBR20-800 mm	2	\$44.90
Belt	GT2 Timing Belt	1	\$16.88

Item	Component	Qty.	Price total
Aluminum Beams	4040 T-Slot Extruded Aluminum Profile, 1000 mm	4	\$79.99
Relay	One-Channel Relay Module	1	\$3.70
Buzzer	Gikfun Passive Electronic Buzzer	2	\$1.80
MOSFET Transistor	IRLZ44N MOSFET Transistors	4	\$4.00
Terminal Block	Glarks Terminal Block Set	2	\$0.26
Flyback Diode	BOJACK 1N4007 Rectifier Diode	1	\$0.05
Wires	22 AWG Stranded Wire Spool	1	\$12.29
Cable Wrap Insulation	Eventronic Heat-Shrink Tubing 2:1	1	\$6.99
Fuse Holder	FEBRYTOLD 0-Gauge Inline Fuse Holder	1	\$4.25
Resistor Kit	Upvivi Resistor Assortment Kit	1	\$6.99
T-Nuts	4040 Series T-Sliding Nuts	15	\$9.99
Bolt and Nut Kit	M5 Button-Head Socket Cap Screw and Nut Kit	1	\$16.99
Motor Mounting	NEMA 23 Steel Mounting Bracket	4	\$11.99
Nut Block	Lysee Lead-Screw Nut Block	2	\$20.19
Conveyor Belt	12 V DC Electric Desktop Conveyor Belt	1	\$99.51
Total without conveyor belt (before tax)			\$525.94
Total with conveyor belt (before tax)			\$625.45

3.2 Proposed Timeline

The proposed implementation schedule spans sixteen weeks. Mechanical construction and firmware development proceed in parallel so that each subsystem can be tested soon after installation. The schedule intentionally reserves the final weeks for integrated testing, debugging, refinement, and presentation preparation.

Legend: **A** Adriano **J** Josué **B** Both

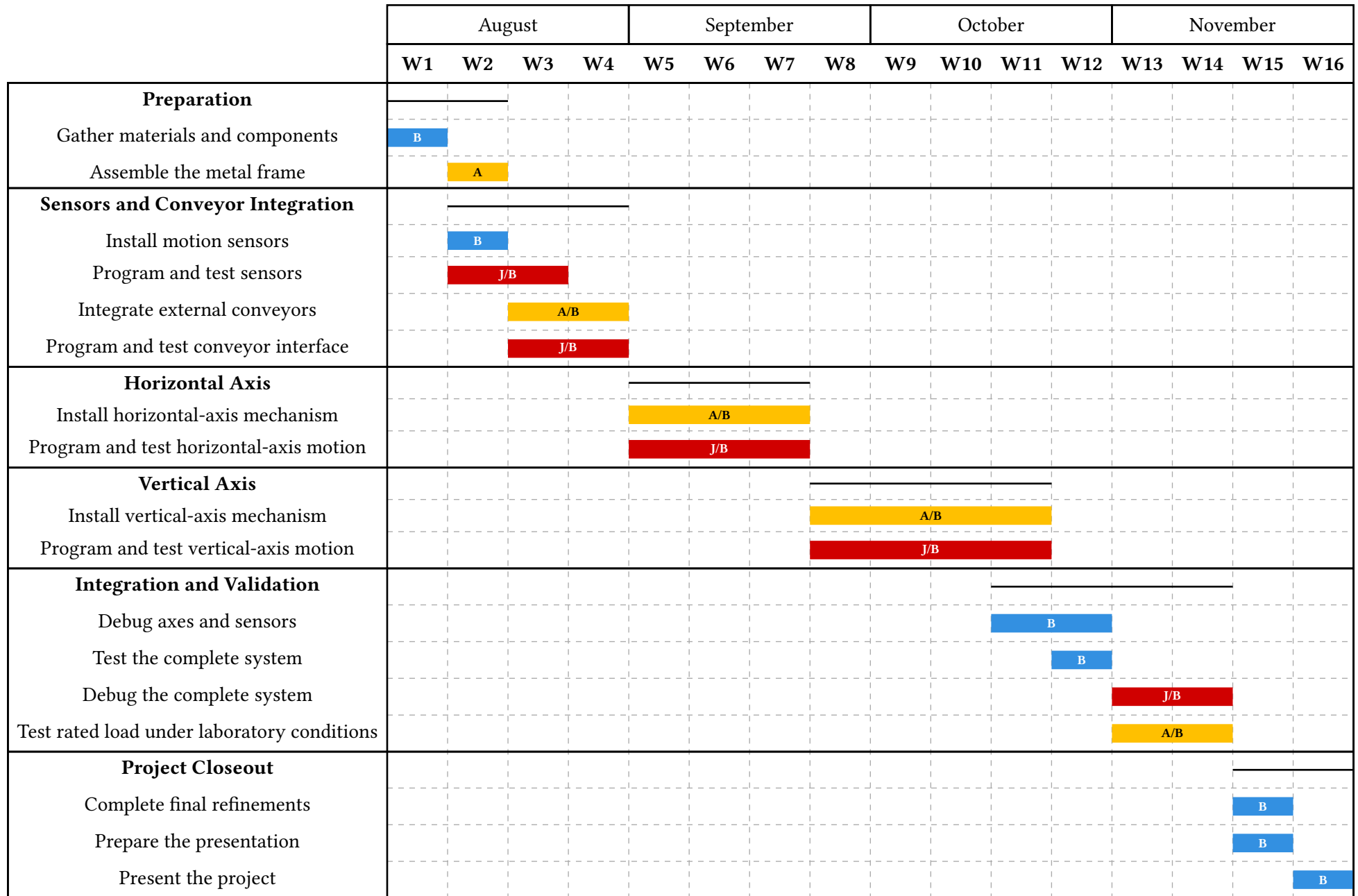


Fig. 3.1: Sixteen-week Safe Dip implementation schedule.

3.3 Task Distribution

The task assignments shown in Fig. 3.1 identify a primary owner for each activity, but they do not prevent either team member from assisting with work outside that assignment. Mechanical and software tasks are deliberately paired during integration so that installation, programming, and validation decisions are made jointly.

Table 3.2
Preliminary distribution of project responsibilities.

Team member	Primary responsibilities
Josué Bouchard	Embedded firmware; time-of-flight and limit-switch interfacing; horizontal- and vertical-axis motion control; supervisory state machine; Modbus TCP integration; system debugging; and software-oriented verification.
Adriano Duque Mena	Mechanical frame assembly; installation of rails, transmissions, motors, sensors, and external conveyor interfaces; mechanical troubleshooting; fabrication support; and preparation of rated-load testing.
Both	Component procurement; electrical and mechanical integration; emergency-stop validation; complete-system testing; final refinement; report preparation; and presentation.

3.4 Environmental Aspects

The laboratory prototype is not intended to use a heated bath or process chemicals. Its direct environmental impact is therefore primarily associated with electricity consumption, packaging waste, metal and plastic fabrication waste, and the eventual disposal of electronic components. Scrap aluminum, steel, wire, and electronic assemblies should be separated whenever practical so they can be reused or sent through an appropriate recycling stream rather than discarded as mixed waste.

The intended industrial application may use wax, resin, or another coating medium at elevated temperature. Because that operating condition is outside the current prototype test envelope, the proposal does not claim validation of spill control, fume generation, coating disposal, or thermal energy consumption. Those subjects must be evaluated using the selected coating material and production process before the system is deployed outside laboratory conditions.

3.5 Health and Safety

Safety is the central motivation for Safe Dip. The system is intended to remove the operator from direct interaction with a repetitive dipping task and, in the full application, from proximity to hot and viscous coating material. The prototype will be evaluated under laboratory conditions without process heat, handling a workpiece of approximately 1 lb. However, the system was designed in preparation for handling 10 kg workpieces, which will present meaningful mechanical, electrical, and suspended-load hazards.

The movement emergency stop is implemented as a Category 0 stop of the movement subsystem. Actuating the normally-closed mushroom button removes power from both stepper-motor drivers immediately. The button requires a twist action to reset, and motion must remain inhibited until the button has been reset and the operator deliberately restarts the system. The electromagnet is powered from a separate supply that is not interrupted by the motion emergency stop so that an attached workpiece is not intentionally released when axis power is removed.

Table 3.3
Primary prototype hazards, controls, and residual limitations.

Hazard	Prototype control	Residual limitation
Unexpected axis motion	Normally-closed, twist-reset mushroom emergency stop removes power from both motor drivers. Firmware prevents automatic restart after a stop event.	The emergency stop does not replace safe setup procedures or physical separation from the moving gantry.
Pinch and crush points	Motion is limited to controlled laboratory tests, and personnel must remain clear of the rails, leadscrew, belt transmission, and carriage while the system is enabled.	The current prototype does not claim the protection provided by a fully guarded industrial cell.

Hazard	Prototype control	Residual limitation
Suspended workpiece	The magnet remains powered during a motion emergency stop. Retention is tested before complete-cycle operation, and no person may stand beneath the workpiece.	A conventional electromagnet does not retain the workpiece after a total loss of magnet power. This is a prototype limitation and must not be described as fail-safe load retention.
Electrical fault or short circuit	Fuses are installed in the prototype power branches, and exposed connections are insulated and secured before operation.	Fuse ratings and conductor sizes must be finalized from the actual branch currents and selected hardware.
Loss of position reference	Normally-closed limit switches provide homing and boundary references, while the time-of-flight sensor verifies workpiece presence before the cycle begins.	Sensor detection does not make entry into the machine workspace safe while motion is enabled.
Thermal or coating exposure	No heated coating material is used during prototype acceptance testing.	The prototype does not validate operation at the intended industrial temperature or with a production coating medium.

The conventional electromagnet creates an important distinction between an emergency stop and a complete power failure. The separate supply preserves holding force when only the motion subsystem is stopped, but a failure of the magnet supply may still release the load. This limitation must remain visible in the operating instructions, test procedure, and final evaluation of the prototype.

3.6 Ethical Aspects

The ethical responsibility of the project team extends beyond demonstrating that the mechanism can complete a pick-dip-place cycle. Because Safe Dip is presented as a safety-oriented system, performance and safety claims must be based on documented tests rather than design intent alone. In particular, the report must clearly distinguish between the intended 10 kg,

elevated-temperature product concept and the laboratory prototype, which is tested without heat and with several safeguards that remain incomplete relative to an industrial installation.

The team will report failed tests, unresolved hazards, and prototype limitations honestly; will not describe the conventional electromagnet as retaining the load after complete power loss; and will not claim compliance with an industrial safety standard unless the relevant architecture and validation have actually been completed. Libraries, technical references, component data, figures, and external designs used during development will be credited appropriately. Safety decisions will take priority over cycle speed, appearance, or demonstration convenience.

Safe Dip does not require the collection of personal information. Operational data communicated through the serial or Modbus TCP interface is limited to machine state, sensor values, commands, and fault information. Any future logging system should collect only the information needed for diagnostics and process traceability.

3.7 Social Aspects

Safe Dip is intended to reduce worker exposure to a repetitive and potentially hazardous manual process. Removing the operator from the immediate dipping area may reduce physical strain and, in the full-temperature application, reduce the risk of burns and contact with hot coating material. A lower-cost gantry system may also make basic automation more accessible to small and medium manufacturers that cannot justify a full industrial robotic cell.

Automation can also create concern about the replacement of manual labor. The proposed system is better understood as transferring the operator away from direct handling and toward setup, supervision, inspection, maintenance, and process control. Its value should therefore be evaluated not only through throughput, but also through changes in worker exposure, training needs, system reliability, and the quality of the resulting work environment.

3.8 Sustainability

The design supports long-term maintenance by using a modular frame and commercially available motors, drivers, sensors, power supplies, and communication hardware. A failed subsystem can therefore be repaired or replaced without discarding the complete machine. The aluminum structure, rails, and mechanical transmission can also remain in service when the controller, sensors, or end effector are upgraded.

The conventional electromagnet is a practical prototype choice, but it consumes power continuously while holding a workpiece. The control system should therefore energize it only when retention is required and should avoid unnecessary idle holding periods. This

energy-use tradeoff should be considered when comparing the prototype with a future electro-permanent or mechanical holding solution.

Repeatable control of immersion depth and dwell time may reduce coating inconsistency and material waste in a future process implementation. That benefit is not demonstrated by the present non-heated prototype and should remain a design objective until it can be measured using the selected coating medium.

Chapter 4

Conclusion

Summary

This chapter concludes the Safe Dip proposal by reviewing the purpose and proposed architecture of the system, identifying the expected contribution of the laboratory prototype, and outlining improvements that may be considered in future development.

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4.1 Summary and Conclusion

Safe Dip is a safety-oriented pick-dip-place system designed to reduce worker exposure to the hazards associated with manual hot-melt coating operations. The proposed design uses a two-axis Cartesian gantry to detect, acquire, transport, dip, and release a workpiece without requiring the operator to remain near the dipping area.

The system is controlled by an STM32 microcontroller that coordinates the stepper-driven horizontal and vertical axes, the time-of-flight sensor, limit switches, electromagnet, emergency-stop input, and communication interfaces. Serial communication provides the baseline operator interface, while Modbus TCP through the W5500 module allows system states, sensor information, and fault conditions to be observed by external equipment.

Safety is a central consideration in the design. A normally-closed, twist-reset emergency-stop button immediately removes power from the movement subsystem, while the conventional electromagnet remains powered from a separate supply so that actuating the movement emergency stop does not intentionally release the suspended workpiece. Fuses are also included in the prototype power branches to reduce the consequences of electrical faults.

The laboratory prototype will be tested without a heated coating bath and will handle a workpiece of approximately 1 lb under controlled conditions. Successful testing would demonstrate the feasibility of the control, sensing, communication, movement, and emergency-stop architecture, but it would not validate operation at the intended industrial temperature. The conventional electromagnet also cannot provide load retention during a complete loss of magnet power, which remains an important prototype limitation.

Overall, Safe Dip proposes a lower-cost and modular alternative to manual dipping and more expensive industrial robotic systems. The project provides a foundation for determining whether the design can later be developed into a guarded, temperature-rated, and industrially deployable machine.

4.2 Suggestions for Future Work

Future development should first address the safeguards required for operation near a heated coating bath. These additions may include a transparent physical enclosure, interlocked access points, independent overtemperature protection, spill containment, and visual and audible status indicators.

The main focus of future improvements would be to harden the system by providing thermal protection to the mechanism. The conventional electromagnet may also be replaced by an electro-permanent magnet, mechanically locking gripper, or another end effector

capable of retaining the workpiece during a complete power failure. Additional sensing could verify that the workpiece has been successfully acquired before movement begins.

Further improvements may include closed-loop position feedback, enhanced fault monitoring, a dedicated operator interface, and standardized communication with upstream and downstream conveyor equipment. After these safeguards are implemented, the system could be evaluated using the intended coating material and operating temperature to measure retention, positioning accuracy, dip consistency, and long-term reliability.

Also, because the electromagnet is deliberately held during an emergency stop, it depends on its 12 V supply remaining available. A planned future improvement is to add a small backup battery on the electromagnet rail, sized to hold the coil energized for a few minutes in the event of a total power loss, so that a suspended part is not dropped when mains power is lost rather than merely when the E-stop is actuated

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Appendix A - Calculations

Leadscrew calculations

The vertical-axis capacity is determined by combining the leadscrew kinematics with the motor's torque-speed characteristic. For a leadscrew with lead L , the required rotational speed is obtained from the desired linear speed using Fig. A.1. The motor's torque-speed curve then provides the available torque $T(\omega)$ at that operating speed. Finally, Fig. A.2 converts the available torque into the maximum safe payload.

$$v = nL$$

Fig. A.1: Relationship between leadscrew rotational speed and linear speed.

$$\begin{aligned} F_{\text{axial}} &= \frac{2\pi T(\omega)\eta}{L} \\ m_{\text{max}} &= \frac{F_{\text{axial}}}{g} = \frac{2\pi T(\omega)\eta}{Lg} \\ m_{\text{safe}} &= \frac{m_{\text{max}}}{S} = \frac{2\pi T(\omega)\eta}{SLg} \end{aligned}$$

Fig. A.2: Maximum safe payload mass produced by the vertical leadscrew.

In these equations, L is the leadscrew lead, defined as the axial distance traveled by the nut during one complete revolution [5]. The term η represents the mechanical efficiency of the leadscrew and accounts for frictional losses, while S is the selected safety factor. For this design, a safety factor of $S = 2$ is applied to account for uncertainties in loading, acceleration, drivetrain losses not captured by the nominal efficiency, and variations between the theoretical and available motor torque.

The torque used in Fig. A.2 should be obtained from the motor's torque-speed curve at the intended operating speed. The static holding torque should not be used for a moving-axis calculation because the available motor torque decreases as rotational speed increases.

Appendix B - Applicable Engineering Codes and Standards

The codes and standards in Table B.1 were identified as relevant references for the design and development of Safe Dip. Their inclusion does not represent a claim of certification or complete compliance. Formal compliance would require evaluation of the finished system against the complete requirements of each applicable document.

Table B.1
Engineering codes and standards applicable to the Safe Dip project.

Code or Standard	Organization	Explanation	Relevance to Safe Dip
IEEE Code of Ethics	Institute of Electrical and Electronics Engineers	Defines ethical duties related to safety, honesty, competence, and public welfare.	Supports safe design, truthful reporting, and clear disclosure of prototype limitations.
NSPE Code of Ethics for Engineers	National Society of Professional Engineers	Requires engineers to prioritize public safety and communicate technical information honestly.	Applies to safety decisions, testing, risk disclosure, and performance claims.
ISO 13850:2015	International Organization for Standardization	Defines requirements and design principles for machinery emergency-stop functions.	Guides the latching E-stop, immediate motion cutoff, manual reset, and restart inhibition.
ISO 14118:2017	International Organization for Standardization	Addresses prevention of unexpected machine start-up.	Requires a deliberate restart action after the E-stop is reset.

Code or Standard	Organization	Explanation	Relevance to Safe Dip
NFPA 79	National Fire Protection Association	Covers electrical equipment, wiring, protection, and controls for industrial machinery.	Guides internal wiring, branch protection, grounding, control circuits, and E-stop wiring.
NFPA 70 National Electrical Code	National Fire Protection Association	Defines requirements for safe electrical installations and supply connections.	Guides power connections, wiring, fuses, grounding, and electrical fault protection.

Appendix C - Meet the Team



Josué Bouchard is originally from Argentina and is completing his Bachelor's degree in Electrical and Computer Engineering Technology at Valencia College. His interest in engineering began with an Arduino board and grew from a fascination with using software to interact with the physical world. After beginning his studies in Computer Engineering at the Universidad de Buenos Aires, he transferred to Valencia College, where he combined his theoretical background with hands-on experience in electronics, microcontrollers, and embedded systems. Josué also enjoys helping others understand technical concepts through his work as a laboratory aide. After graduation, he hopes to pursue a career in embedded systems and eventually work in aerospace or robotics.



Adriano Mena Duque de Estrada was born in Colón, Matanzas, Cuba, and immigrated to the United States with his family in 2016, crossing the border with very little and no knowledge of English. With faith and perseverance, he is now completing his Bachelor's degree in Electrical and Computer Engineering Technology at Valencia College, having switched from Aerospace Engineering after realizing it wasn't the right fit — though designing spacecraft remains a personal dream. Outside of coursework, Adriano enjoys strategy video games, spends time with his dog and bearded dragon, and loves building electrical projects, which is what ultimately drew him into this career path.